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ORIGINAL ARTICLE

Modeling SARS-CoV-2 Infection Dynamics: Insights into Viral Clearance and Immune Synergy

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Abstract

Understanding the mechanisms of interaction between SARS-CoV-2 infection and the immune system is crucial for developing effective treatment strategies against COVID-19. In this paper, a mathematical model is formulated to investigate the interactions among SARS-CoV-2 infection, cellular immunity, and humoral immunity. Clinical data from eight asymptomatic or mild COVID-19 patients in Munich are used to fit the model, and the dynamics of natural killer (NK) cells, cytotoxic T lymphocytes (CTLs), B cells, and antibodies are further explored using the average of the best-fitting parameter values. Subsequently, the impact of NK cells, CTLs, B cells, and antibodies on SARS-CoV-2 infection is numerically investigated. The results indicate that (i) the synergy of NK cells, CTLs, and antibodies leads to a rapid decrease in the viral load during SARS-CoV-2 infection; (ii) antibodies play a crucial role compared to other immune mechanisms, and enhancing B cell stimulation may be more effective in clearing the virus from the lungs; (iii) in terms of cytotoxic effects, CTLs are stronger and more sustained than NK cells. Furthermore, the existence and local stability of the model's equilibria are fully classified, and complex dynamics of the model are further investigated using bifurcation theory, revealing multistability phenomena, including multiple attractors and periodic solutions. These findings suggest potential uncertainty and diversity in SARS-CoV-2 infection outcomes.

Keywords SARS-CoV-2 infection · Immune synergy · Mathematical model · Viral dynamics

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1 Introduction

The COVID-19 pandemic caused by SARS-CoV-2 (Tay et al. 2020), which emerged at the end of 2019, is the most significant epidemic catastrophe that humanity has encountered in the last hundred years. Over the past three years, according to incomplete statistics, COVID-19 has infected approximately 770 million people worldwide, claimed over 7 million lives (World Health Organization 2023), and pushed the world economy into the most severe recession (Sumner et al. 2020). Despite the mitigation of COVID-19, it still remains a significant concern due to the ongoing evolution of new variants (Korosec 2023; Flores-Vega et al. 2022; Dubey et al. 2022; Telenti et al. 2022; Markov et al. 2023; Phan et al. 2024). Therefore, it is crucial to understand SARS-CoV-2 infection and uncover the underlying factors behind the clinical data.

Mathematical models are valuable explanatory and supplementary tools for studying the dynamics of SARS-CoV-2 infection (Hernandez-Vargas and Velasco-Hernandez 2020; Gonçalves et al. 2020; Dobrovolny 2020; Schuh et al. 2024; Yang et al. 2024; Korosec et al. 2023), complementing insights from experimental and clinical research. Perelson and Ke (2021) provided an overview of the application of simple target cell models in various viral infections and treatments. They point out that more complex models, which have evolved from these simple models, can capture more of the virology and immunology involved (Getz et al. 2020; Sego et al. 2020; Voutouri et al. 2021). The innate immune response, serving as the first defense against viral infections, may influence the kinetics of SARS-CoV-2 infection. Goyal et al. (2020) used a density-dependent rate in the infected cells equation to model the killing effect on infected cells by innate mechanisms. The results indicate that this early response may result in the quick elimination of millions of infected cells. Sanche et al. (2022) incorporated resting innate immune cells and activated immune cells into a within-host COVID-19 infection model, and the simulation results suggest that specifically targeting aspects of the innate immune response may prove beneficial compared to broadly acting anti-inflammatory agents.

Cytotoxic T lymphocytes (CTLs) are a critical component of the adaptive immune system and play an important role in immune defense against intracellular pathogens. CTL-mediated adaptive immune responses have been incorporated into SARS-CoV-2 infection models. Hernandez-Vargas and Velasco-Hernandez (2020) adapted a minimalistic model including the T cell immune response and suggested that it is the best model to fit the data from Wölfel et al. (2020). Goyal et al. (2020) incorporated the differentiation of effector cells into a COVID-19 model to capture the coupled interactions of susceptible cells, infected cells, SARS-CoV-2, and a mounting immune response. The results suggest that the acquired immune response may induce the eradication of infected cells in the upper airway. Wang et al. (2020) formulated an ODE model that uses an exponential function to represent the clearance of infected cells by the adaptive immune response. Data fitting and numerical simulations suggest that the viral dynamics of SARS-CoV-2 infection consist of three phases, with the second plateau phase likely attributed to the presence of possible secondary target cells.

Humoral immunity is another adaptive immune response that aims to achieve protection through the production of antibodies provided by B cells. Because antibodies bind to critical sites on the free virus, preventing it from infecting target cells (Moderbacher et al. 2020; Suthar et al. 2020; Long et al. 2020), they provide protective immunity (Fraussen 2022). Investigating the impact of humoral immunity on the dynamics of SARS-CoV-2 infection has been a significant focus in recent years. Elaiw et al. (2024) incorporated humoral immunity into a SARS-CoV-2 infection model, and the simulation results show that vigorous activation of humoral immunity can suppress viral replication. Costa et al. (2023) proposed a high-dimensional ODE system that includes cellular immunity, humoral immunity, and proinflammatory cytokines, and demonstrated that the model can capture cohort data on $CD4^{+}$, $CD8^{+}$, and viremia behavior in severe cases of COVID-19.

The studies mentioned above have played an important role in helping unravel the mechanisms of interaction between viral dynamics and the immune system, contributing to the development of treatment strategies against COVID-19. These mathematical models consider the effects of one or two immune mechanisms on SARS-CoV-2 infection. However, natural killer (NK) cells, CTLs, and the humoral immune response are the three main mechanisms of immunity against SARS-CoV-2 infection. Therefore, in this paper we will incorporate NK cells, CTLs, and humoral immunity into a SARS-CoV-2 infection model. The purpose of the paper is to utilize the model, combined with some data, to improve our understanding of the interaction between SARS-CoV-2 infection and the immune system, as well as to further evaluate the contributions of NK cells, CTLs, and humoral immunity in combating SARS-CoV-2 infection.

This paper is organized as follows. In Sect. 2, based on the biological reviews of SARS-CoV-2 infection, we propose a seven-dimensional ODE model to describe the interaction between SARS-CoV-2 infection and three fundamental mechanisms of the immune system. In Sect. 3, clinical data from eight patients with COVID-19 are used to fit the model, and the viral dynamics of the model are investigated using the average values of the best-fitting parameters. In Sect. 4, numerical investigations are conducted to explore the impact of NK cells, CTLs, and antibodies secreted by B cells on SARS-CoV-2 infection. In Sect. 5, we analyze the existence and local stability of equilibria and further investigate the complex dynamics of the high-dimensional system through bifurcation analysis and simulation. A brief discussion is provided in Sect. 6.

2 Model Formulation

In this section, we formulate a 7-dimensional ordinary differential equation model to describe the dynamic interactions among SARS-CoV-2 infection, cellular immunity, and humoral immunity.

A brief review of the immune system's response to SARS-CoV-2 will be useful. The exact target of SARS-CoV-2 infection is not fully understood (Wang et al. 2020). Based on findings related to SARS-CoV-2 infection (Xu et al. 2020; Kuster et al. 2020; Baig et al. 2020), after entering the host through the respiratory tract, type 2 alveolar epithelial (T2AE) cells are likely the primary target (Wang et al. 2020; Haagmans et al. 2004; Harrison et al. 2020), as these cells exhibit high expression of the SARS-CoV-2 receptor, angiotensin-converting enzyme 2 (ACE2), and other proviral genes necessary for productive viral replication.

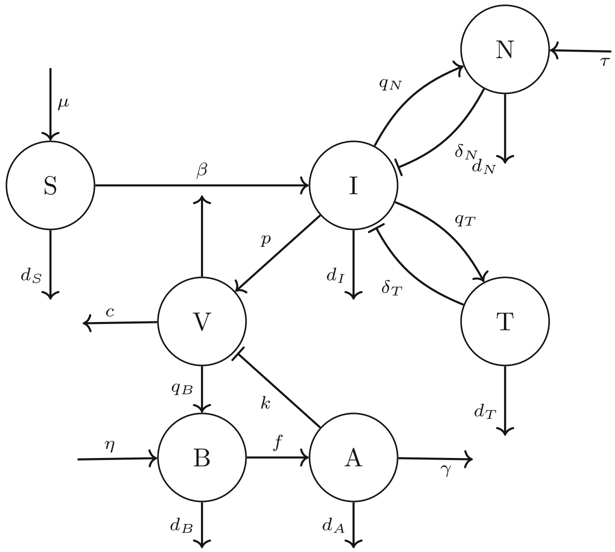


Fig. 1 The schematic diagram illustrates the interactions among SARS-CoV-2 infection, NK cells, CTLs, B cells, and antibodies. S denotes susceptible cells; I, infected cells; N, NK cells; T, CTLs; V, viral load; B, B cells; and A, antibodies. All the variables and parameters are summarized in Table 1

According to the World Health Organization (WHO) (2022), when SARS-CoV-2 enters the body, NK cells recognize markers present on the virus and are activated to remove the virus. NK cells have the significant ability to directly target infected cells without the need for specific antigen presentation (Masselli et al. 2020), inducing apoptosis of infected cells through various pathways (Ahmed et al. 2020). Subsequently, the adaptive immune response is initiated. CTLs, B cells, and antibodies are crucial components of adaptive immunity. CTLs induce significant apoptosis in infected cells (Toor et al. 2021; Groscurth 1989). Antibodies, also referred to as immunoglobulins, are secreted by plasma cells, which differentiate from B cells. They circulate in the bloodstream and tissues, where they bind to foreign pathogens and mitigate their detrimental effects (Vojdani et al. 2020).

Based on the biological review of SARS-CoV-2 infection, the population of target cells is divided into susceptible cells and infected cells, represented by S and I respectively, for their concentrations within the lung. The concentration of the viral load of SARS-CoV-2 is denoted by V. The concentrations of NK cells and CTLs in the lung are represented by N and T, respectively. Humoral immunity provides protection through antibodies, denoted by A, which are secreted by B cells, with the concentration of B cells denoted by B. The schematic diagram illustrating the interactions among SARS-CoV-2 infection, cellular immunity, and humoral immunity is shown in Fig. 1. Based on the transition diagram in Fig. 1, the model is described by the following system of differential equations:

$$
\begin{cases}
\dfrac{\mathrm{d}S}{\mathrm{d}t} = \mu - d_S S - \beta V S, \\
\dfrac{\mathrm{d}I}{\mathrm{d}t} = \beta V S - \delta_N I N - \delta_T I T - d_I I, \\
\dfrac{\mathrm{d}V}{\mathrm{d}t} = pI - cV - kVA, \\
\dfrac{\mathrm{d}N}{\mathrm{d}t} = \tau + q_N I - d_N N, \\
\dfrac{\mathrm{d}T}{\mathrm{d}t} = q_T I T - d_T T, \\
\dfrac{\mathrm{d}B}{\mathrm{d}t} = \eta + q_B B V - d_B B - e B^2, \\
\dfrac{\mathrm{d}A}{\mathrm{d}t} = f B - \gamma A V - d_A A.
\end{cases}
\tag{1}
$$

In System (1), susceptible cells (S) are produced at a constant rate μ and die at a rate d_S. They are infected by SARS-CoV-2 at a rate of $\beta V S$, where β is the infection rate. Infected cells (I) are assumed to be killed by NK cells at a rate δ_N (Ahmed et al. 2020) and by CTLs at a rate δ_T (Andersen et al. 2006; Groscurth 1989), and they decay at a rate d_I. The virus (V) replicates within infected cells, bursts out at a rate p, and decays at a rate c. The virus is neutralized by antibodies at a rate k. Since healthy individuals maintain a consistent level of lymphocytes (Wang et al. 2020), and both NK cells and B cells can be recruited to the infection site in the early stage of infection under the influence of chemokines (Sokol and Luster 2015), we assume that NK cells (N) and B cells (B) have constant recruitment rates, denoted by τ and η, respectively. As the generic innate immune response factor is produced at a rate proportional to the concentration of productively infected cells and decays at a constant rate (Best et al. 2021), NK cells are stimulated by infected cells at a rate of q_N and decay at a rate d_N. We also assume that B cells are stimulated by virus particles at a rate q_B and die at a rate d_B, as described in (Gujarati and Ambika 2014). The density-dependent factor of B cells, represented by e, negatively regulates B cell production, limiting excessive growth (Brauer and Castillo-Chavez 2012). Antibodies (A) are secreted by B cells at a rate f, are lost in the formation of antibody-virus complexes at a rate γ (Suthar et al. 2020), and naturally degrade at a rate d_A. We assume that CTLs (T) are stimulated by infected cells at a rate q_T and decay at a rate d_T. All parameters are non-negative, and the definitions of all variables and parameters are summarized in Table 1.

The purpose of this paper is to use Model (1) to investigate the dynamics among SARS-CoV-2 infection, cellular immunity and humoral immunity. It also aims to evaluate the impact of NK cells, CTLs, B cells, and antibodies on SARS-CoV-2 infection, thereby enhancing our understanding of the interaction mechanism between SARS-CoV-2 infection and the immune system from a mathematical perspective.

Table 1 Parameters and variables

Variables/parameters	Description	Value	Refs.
S	Susceptible cells	–	–
I	Infected cells	–	–
V	The virus	–	–
N	NK cells	–	–
T	CTLs	–	–
B	B cells	–	–
A	Antibodies	–	–
$S(0)$	Initial number of susceptible cells	6×10^4 cells ml^{-1}	Wang et al. (2020)
$I(0)$	Initial number of infected cells	0 cells ml^{-1}	Wang et al. (2020)
$V(0)$	Initial number of the virus	10^{-3} RNA copies ml^{-1}	Wang et al. (2020)
$N(0)$	Initial number of NK cells	100 cells ml^{-1}	Assumed
$T(0)$	Initial number of CTLs	10^{-3} cells ml^{-1}	Assumed
$B(0)$	Initial number of B cells	130 cells ml^{-1}	Assumed
$A(0)$	Initial number of antibodies	0 ng ml^{-1}	Assumed
μ	Productivity of susceptible cells	420 cells ml^{-1} day^{-1}	Estimated
d_S	Base death rate of susceptible cells	0.007 day^{-1}	Barkauskas et al. (2013); Desai et al. (2014)
d_I	Base death rate of infected cells	0.47 day^{-1}	Dobrovolny et al. (2013)
c	Clearance rate of the virus	10 day^{-1}	Ke et al. (2021)
τ	Recruitment rate of NK cells	1×10^3 cells ml^{-1} day^{-1}	Assumed
d_N	Death rate of NK cells	0.07 day^{-1}	Zhang et al. (2007)
d_T	Death rate of CTLs	0.05 day^{-1}	Estimated
η	Recruitment rate of B cells	2×10^3 cells ml^{-1} day^{-1}	Assumed

Table 1 continued

Variables/parameters	Description	Value	Refs.
d_B	Death rate of B cells	$0.033\ \text{day}^{-1}$	Fulcher and Basten (1997)
d_A	Degradation rate of free antibodies	$0.12\ \text{day}^{-1}$	Estimated
β	Viral infection rate	1.639×10^{-6} ml day^{-1}	Fitted
δ_N	Killing rate by NK cells on infected cells	$3.304 \times 10^{-7}\ \text{day}^{-1}$	Fitted
δ_T	Killing rate by CTLs on infected cells	$1.217 \times 10^{-5}\ \text{day}^{-1}$	Fitted
p	Viral production rate	$1848\ \text{day}^{-1}$	Fitted
k	Neutralization rate by antibodies	3.999×10^{-4} copies day^{-1}	Fitted
q_N	Activation rate of NK cells	$0.461\ \text{day}^{-1}$	Fitted
q_T	Activation rate of CTLs	$1.372 \times 10^{-4}\ \text{day}^{-1}$	Fitted
q_B	Rate at which B cells are stimulated by virus particles	4.884×10^{-7} cells day^{-1}	Fitted
e	Density-dependent factor of B cells	1.078×10^{-9}ml $\text{cells}^{-1}\ \text{day}^{-1}$	Fitted
f	Production rate of antibodies by B cells	$0.050\ \text{day}^{-1}$	Fitted
γ	The loss rate of free antibodies in the formation of antibody-virus complexes	$6.081 \times 10^{-7}\ \text{day}^{-1}$	Fitted

3 Data Fitting

The viral load data used in this paper were obtained from the first batch of COVID-19 patients in Munich, Germany (Wölfel et al. 2020). All patients were tested for viral RNA levels using reverse transcription polymerase chain reaction (RT-PCR) on a daily basis after the onset of symptoms. In this study, we selected data collected exclusively from sputum samples, as done in other studies on SARS-CoV-2 (Goyal et al. 2020; Wang et al. 2020; Ke et al. 2021). The reason for selecting this dataset is that all patients were asymptomatic or had mild cases, experiencing self-immune recovery without treatment.

Due to limited data, we begin by fixing some parameter values based on existing literature and biological significance. According to Wang et al. (2020), the total number of alveolar cells in the human body is approximately 6×10^8 cells, with T2AE cells constituting 60% of the total. Given a lung volume of 6000 ml Ph (1993), the steady-state value of susceptible cells can be estimated as 6×10^4 cells ml^{-1}, which is used as the initial value of susceptible cells $S(0)$. The death rate of uninfected cells d_S is chosen as 0.007 day^{-1}, since alveolar cells have a lifespan of 100 $\sim$ 200 days (Barkauskas et al. 2013; Desai et al. 2014). Thus, the production rate of susceptible cells (μ) is calculated as $\mu = d_S S(0) = 420$ cells ml^{-1} day^{-1}. The initial number of infected cells $I(0)$ is postulated to be 0 cells ml^{-1}.

Since both SARS-CoV-2 and influenza A virus are acute respiratory infectious diseases, the base death rate of infected cells (d_I) is assumed to be 0.47 day^{-1} (Dobrovolny et al. 2013). The value of d_I is within the range found in many studies that fit the SARS-CoV-2 infection model to clinical data from patients with COVID-19 (Gonçalves et al. 2020; Kim et al. 2021; Iwanami et al. 2020). We assume that the effective initial viral load $V(0)$ is 10^{-3} RNA copies ml^{-1}, as used in Best et al. (2017), Ke et al. (2021), Wang et al. (2020), to account for the early lung infection where only a small proportion of viruses are transported to the fluid immediately after infection. The clearance rate of the virus (c) is hypothesized to be 10 day^{-1}, as prescribed in another study on the in vivo kinetics of SARS-CoV-2 infection (Ke et al. 2021).

The initial number of NK cells $N(0)$ and B cells $B(0)$ are assumed to be 100 cells ml^{-1} and 130 cells ml^{-1}, respectively, as their concentration in the lung is much smaller than that of T2AE cells. Since the recruitment rate of lymphocytes to the infection site is 10^4 cells ml^{-1} day^{-1} (Wang et al. 2020), and NK cells account for approximately 10%$\sim$20% (Zhang et al. 2007), while B cells account for about 20%$\sim$40% (Cooper and Alder 2006) of peripheral blood lymphocytes at any given time, we assume that the recruitment rates of NK cells (τ) and B cells (η) are 1×10^3 cells ml^{-1} day^{-1} and 2×10^3 cells ml^{-1} day^{-1}, respectively. The death rate of NK cells (d_N) is estimated to be 0.07 cells day^{-1}, based on the circulating half-life of mature NK cells being approximately 7 $\sim$ 10 days (Zhang et al. 2007). The death rate of B cells (d_B) is fixed at 0.033 cells day^{-1}, as the lifespan of B cells is about 4.3 weeks (Fulcher and Basten 1997).

Since CTLs must differentiate from naive CD8$^+$ T cells before proliferating (Wang et al. 2000), the initial value of CTLs $T(0)$ is postulated to be 0 cells ml^{-1} and set to 0.001 cells ml^{-1} for mathematical simulation, as used in Wang et al. (2020). Given that the half-life of CTLs ranges from several weeks to months, the natural death

Table 2 Comparison of models with different immune mechanisms that influence the dynamics of viral infection

Model	Basic model	NK cells	CTLs	B cells	Antibodies	Fitting results	Parameter values
(A1)	Yes	No	No	No	No	Fig. 14	Table 4
(A2)	Yes	Yes	No	No	No	Fig. 15	Table 5
(A3)	Yes	Yes	Yes	No	No	Fig. 16	Table 6
(A4)	Yes	Yes	No	Yes	Yes	Fig. 17	Table 7
(1)	Yes	Yes	Yes	Yes	Yes	Fig. 2	Table 3

rate of CTLs is hypothesized to be 0.05 cells day^{-1}. The initial number of antibodies $A(0)$ is fixed at 0 ng ml^{-1}. In the early stages of COVID-19 infection of the mucosal surfaces, immunoglobulin A (IgA) predominates in the humoral immune response specific to SARS-CoV-2 (Sterlin et al. 2021; Takamatsu et al. 2022). According to Iii et al. (2019), the half-life of IgA is estimated to be between 5 and 7 days (and it could be longer). Therefore, we assumed the decay rate of antibodies to be 0.12 day^{-1}. All the above parameter values are summarized in Table 1.

To compare the dynamics of viral infection with and without immunity, we consider the following five models. The first model is a basic virus model that includes only susceptible cells, infected cells, and the virus, denoted as Model (A1). Next, NK cells are added to Model (A1), resulting in the second model, denoted as Model (A2). In the third model, both NK cells and CTLs are included in Model (A1), leading to Model (A3). The fourth model incorporates NK cells, B cells, and antibodies to Model (A1), referred to as Model (A4). These four models are presented in Appendix A. The fifth model is Model (1), which we proposed in Sect. 2, integrating both NK cells, CTLs, and humoral immunity. The differences between the five models are summarized in Table 2.

Clinical data from Munich were first used to fit Models (A1), (A2), (A3), and (A4). The best-fitting results are shown in Appendix A (see Figs. 14, 15, 16, 17), and the estimated parameter values are also presented (see Tables 4, 5, 6, 7). The best-fitting results of Model (1) are shown in Fig. 2, and the estimated parameter values are presented in Table 3. The average of the best-fitting parameter values is also summarized in Table 1. From Figs. 14, 15, and 16, it can be seen that the viral load always remains above the RT-PCR detection limit in the lungs of each patient. This suggests that Models (A1), (A2), and (A3) cannot explain the biological phenomenon where the viral load decreases below the RT-PCR detection limit in a short period of time. As illustrated in Fig. 17, the viral load of each patient approaches the RT-PCR detection limit before subsequently rebounding. This result suggests that Model (A4), which incorporates NK cells and humoral immunity, effectively reduces the viral load to near the RT-PCR detection limit within a short period. However, an uncontrolled rebound may still occur.

From Fig. 2, Model (1) effectively captures the process in which the viral load rises to a peak, then declines, and drops below the RT-PCR detection limit in each patient. This suggests that Model (1) may offer some insights into understanding the mechanisms of interaction between SARS-CoV-2 infection and the immune system.

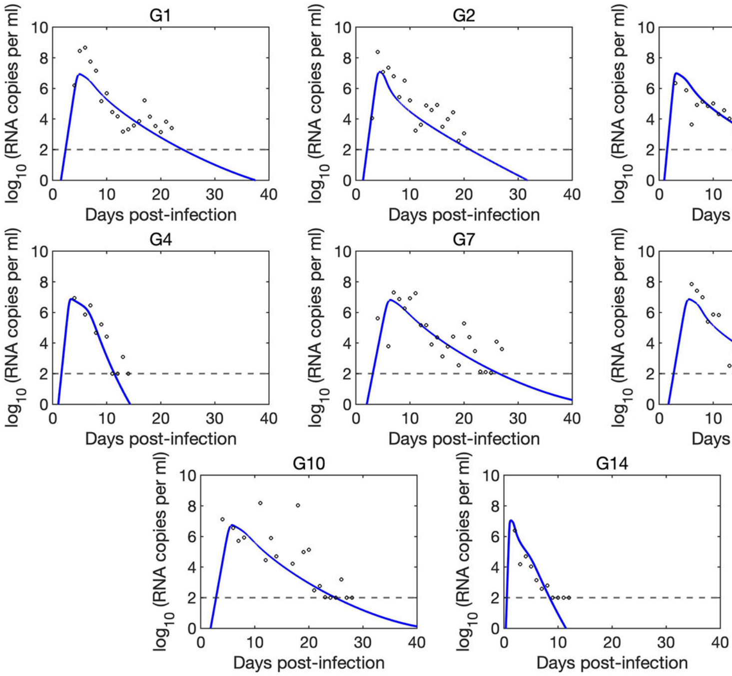


Fig. 2 Best-fitting results of each patients with model (1). The black circles are clinical data, and blue lines are model predictions. The black dashed line represents the RT-PCR detection limit (Color figure online)

Using the average of the best-fitting parameter values, we numerically simulate Model (1), with the time series of Model (1) shown in Fig. 3. From Fig. 3(a) and (b), we observe that the number of susceptible cells rapidly decreases from an initial count of 6×10^4 cells ml^{-1} to a minimum on days 3 to 4, while the viral load rises sharply, peaking on day 3. As the infection progresses, cellular and humoral immunity are activated, leading to control of the infection. Meanwhile, susceptible cells gradually increase, and the viral load drops below the RT-PCR detection limit in the lung between days 16 and 17.

From Fig. 3(c) and (d), the concentration of NK cells increases slowly during days 1 to 3. As the number of infected cells increases significantly, NK cells are rapidly stimulated to proliferate, reaching a peak between days 7 and 8. Subsequently, as infected cells are cleared, the concentration of NK cells gradually decreases. CTLs are activated between days 4 and 5, undergoing rapid and extensive proliferation, peaking between days 11 and 12, and then subsequently declining.

From Fig. 3(e) and (f), B cells undergo rapid proliferation 2 to 3 days after infection, peaking around day 8. Antibodies are produced in significant quantities approximately 4 to 5 days post-infection, reaching their peak around days 18 to 21. This finding aligns with Sterlin et al. (2021), which reports that IgA serum concentrations peak approximately three weeks after symptom onset. Subsequently, as the infection is controlled, both B cell and antibody levels decline.

Table 3 Best-fitting parameter values of Model (1) to the viral load data

Patient	β(ml/virus/day)	δ_N(/day)	δ_T(/day)	p(/day)	k(copies/day)	q_N(/day)
#1	7.926×10^{-7}	3.459×10^{-7}	9.163×10^{-6}	1896	1.079×10^{-4}	0.333
#2	6.892×10^{-7}	1.009×10^{-8}	1.000×10^{-5}	2623	1.989×10^{-3}	0.598
#3	1.404×10^{-6}	1.023×10^{-7}	1.002×10^{-5}	2000	3.267×10^{-4}	0.100
#4	1.580×10^{-6}	7.390×10^{-7}	1.502×10^{-5}	1564	7.057×10^{-5}	0.599
#7	6.991×10^{-7}	5.987×10^{-7}	3.001×10^{-5}	1558	1.841×10^{-4}	0.599
#8	7.886×10^{-7}	2.625×10^{-8}	7.093×10^{-6}	1654	1.040×10^{-5}	0.350
#10	9.590×10^{-7}	1.219×10^{-7}	6.006×10^{-6}	1270	1.040×10^{-5}	0.582
#14	6.197×10^{-6}	6.991×10^{-7}	1.005×10^{-5}	2222	5.000×10^{-4}	0.525
Mean	1.639×10^{-6}	3.304×10^{-7}	1.217×10^{-5}	1848	3.999×10^{-4}	0.461
SD	1.872×10^{-6}	3.084×10^{-7}	7.681×10^{-6}	431.3	6.638×10^{-3}	0.183

Patient	q_T(/day)	q_B(cells/day)	e(ml/cells/day)	f(/day)	γ(/day)
#1	1.291×10^{-4}	3.921×10^{-7}	8.593×10^{-10}	0.013	3.895×10^{-8}
#2	1.100×10^{-4}	3.800×10^{-7}	9.300×10^{-10}	0.010	2.993×10^{-7}
#3	1.200×10^{-4}	3.800×10^{-7}	9.300×10^{-10}	0.014	3.030×10^{-7}
#4	1.800×10^{-4}	4.201×10^{-7}	1.718×10^{-9}	0.010	3.980×10^{-6}
#7	1.230×10^{-4}	3.800×10^{-7}	1.795×10^{-9}	0.010	3.040×10^{-8}
#8	1.280×10^{-4}	6.800×10^{-7}	2.970×10^{-10}	0.011	1.010×10^{-7}
#10	1.350×10^{-4}	6.800×10^{-7}	3.101×10^{-10}	0.010	1.219×10^{-8}
#14	1.726×10^{-4}	5.950×10^{-7}	1.781×10^{-9}	0.321	1.001×10^{-7}
Mean	1.372×10^{-4}	4.884×10^{-7}	1.078×10^{-9}	0.050	6.081×10^{-7}
SD	2.529×10^{-5}	1.383×10^{-7}	6.221×10^{-10}	0.110	1.367×10^{-6}

The dynamics of target cell populations suggest that the maximum percentage of target cells lost reached 95.2% around 10 to 11 days post-infection, as illustrated in Fig. 4. Precise medical data on the percentage of type 2 alveolar epithelial (T2AE) cell loss in humans is lacking. However, findings from Chait et al. (2022) indicate a significant loss of T2AE cells and increased perialveolar lymphocyte cytotoxicity in all fatal COVID-19 cases, even in the early stages before typical patterns of acute lung injury become histologically apparent. It is noteworthy that 75% (6 out of 8) of the selected datasets provided evidence of lung dysfunction (Böhmer et al. 2020). The exact extent of lung tissue loss requires further biological experimentation for validation.

4 Numerical Investigation

In this section, we numerically explore the contributions of NK cells, CTLs, B cells, and antibodies in combating SARS-CoV-2 infection.

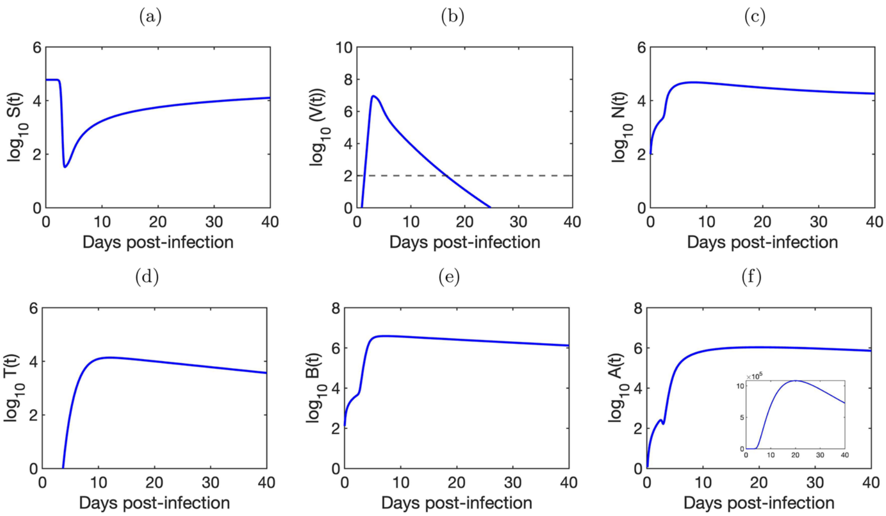


Fig. 3 Predicted dynamics of model (1). The parameter values are fixed in Table 1. The black dashed line in (**b**) represents the RT-PCR detection limit (Color figure online)

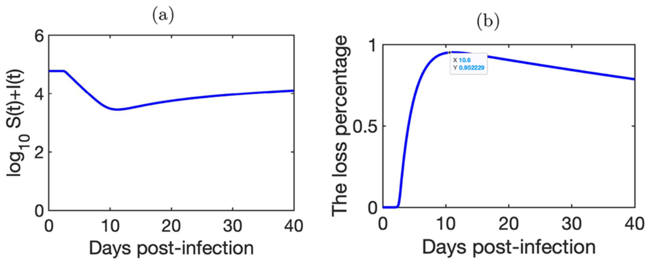


Fig. 4 The dynamics of target cell population. **a** The dynamics of $S(t) + I(t)$. **b** The percentage of target cell lost. The maximum percentage of target cells lost reached 95.2% around 10 to 11 days post-infection (Color figure online)

4.1 Dynamics of SARS-CoV-2 Infection

To illustrate the effects of NK cells, CTLs, B cells, and antibodies on viral clearance, we first investigate the dynamic behaviors of susceptible cells, infected cells, and viral load using Models (A1), (A2), (A3), (A4), and Model (1). By applying the average of the best-fitting parameter values for each patient across the models, we numerically simulate the five models, as shown in Fig. 5. From Fig. 5a, we observe that the number of susceptible cells sharply decreases to its lowest level between days 2 and 4 across all five models. In Models (A4) and (1), which incorporate B cells and antibodies, the rate and magnitude of the increase in susceptible cells are approximately the same. Conversely, in models without B cells and antibodies (Models (A1), (A2), and (A3)), susceptible cell numbers also increase, but the rise is less pronounced than in Models (A4) and (1), ultimately leading to fluctuations.

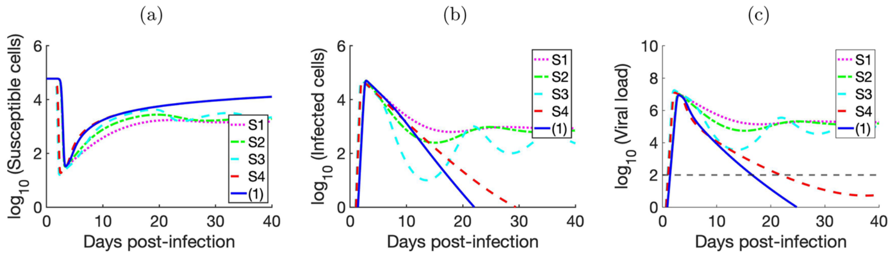


Fig. 5 The prediction of Model (A1), (A2), (A3), (A4), and Model (1): **a** susceptible cells; **b** infected cells; and **c** viral load. All parameter values are based on the average of the best-fitting parameter values for each model. The black dashed line in (**c**) represents the RT-PCR detection limit (Color figure online)

Figure 5b and c show that infected cells and viral load sharply increase within 3 days post-infection in all five models. In the case of (A2), the number of infected cells and viral load decreases more rapidly than in Model (A1), eventually stabilizing at a high level. Model (A3) shows a faster decline in infected cells and viral load compared to both (A1) and (A2). After the viral load drops to a certain level, significant oscillations occur on days 13–14, and it remains at a high level in the case of (A3). This suggests that NK cells and CTLs alone are not sufficient to clear the virus from the body. In the case of (A4), which includes NK cells, B cells, and antibodies, it can be observed that the count of infected cells decreases to nearly zero; however, this process takes longer than in Model (1). The viral load can be reduced below the RT-PCR detection limit, but it also takes longer than in Model (1). Additionally, it appears that the viral load may rebound subsequently.

The presence of NK cells, CTLs, and humoral immunity can significantly reduce the viral load to below the RT-PCR detection limit, potentially down to zero, as demonstrated by Model (1). Furthermore, this reduction in viral load does not exhibit a rapid rebound over a short period. This suggests that the model with added CTLs can effectively eliminate the virus compared to Model (A4). These results show that the synergy of NK cells, CTLs, B cells, and antibodies effectively diminish the viral load in SARS-CoV-2 infections to near zero, with no short-term rebound observed.

4.2 Comparison of Different Immune Mechanisms

To investigate the contributions of NK cells, CTLs, B cells, and antibodies to viral clearance, we simulate Model (1) based on the parameter values given in Table 1, by setting the killing rate by NK cells $\delta_N = 0$, the killing rate by CTLs $\delta_T = 0$, or the neutralization rate by antibodies $k = 0$, respectively. The time series of viral load is shown in Fig. 6. In Fig. 6, the solid line A represents the time series of viral load for Model (1) with all three immune mechanisms. The dotted lines B, C, and D represent the time series of viral load for Model (1) with only NK cells, CTLs, and antibodies, respectively, while the dotted and dashed lines E, F, and G represent the time series of viral load for Model (1) without antibodies, CTLs, and NK cells, respectively.

From Fig. 6, it shows that the viral load could not decrease below the RT-PCR detection limit in cases where $k = 0$, corresponding to lines B, C, and E in Fig. 6.

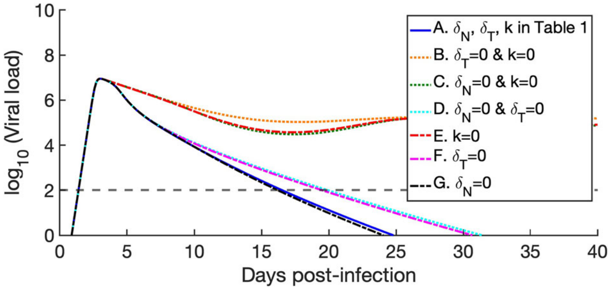


Fig. 6 The contribution of NK cells, CTLs, and antibodies to the clearance of SARS-CoV-2 infection. The black dashed line represents the RT-PCR detection limit (Color figure online)

In contrast, it could decrease below the RT-PCR detection limit in cases involving neutralization by antibodies, which correspond to lines A, D, F, and G. This indicates that antibodies play a more crucial role compared to other immune mechanisms. In both scenarios A and G, which involve CTLs, the time required for the viral load to drop to the detection limit is shorter than in scenarios D and F, which do not include CTLs. This suggests a synergy between the antibody and other mechanisms. Interestingly, by comparing cases A and G, we find that viral load clearance is slightly faster in the absence of NK cells than in their presence. A possible explanation for this phenomenon is that the absence of NK cells may lead to an increase in infected cells and viral load, which in turn enhances the immune effects of CTLs, B cells and antibodies, thereby accelerating the clearance of SARS-CoV-2.

Given the crucial role of antibodies in clearing SARS-CoV-2, we investigated the effect of increased antibody production on the dynamics of SARS-CoV-2 infection. Since antibodies are produced by B cells, we increased the number of antibodies by raising the stimulation rate of B cells, thereby examining the impact of enhanced antibody production on viral infection. In this study, the stimulation rate of B cells, q_B, was selected from three different values: $q_B = 4.884 \times 10^{-7}$, 4.884×10^{-6}, and 4.884×10^{-5}. We then used these three values of q_B to simulate Model (1), and the simulation results are presented in Fig. 7. As shown in Fig. 7e and f, an increase in B cell concentration is accompanied by a significant rise in antibody concentration.

From Fig. 7a, the concentration of susceptible cells begins to decrease between days 2 and 3 after infection in all three scenarios, with a consistent rate of decrease. When $q_B = 4.884 \times 10^{-7}$, the concentration of susceptible cells drops to its lowest level. At $q_B = 4.884 \times 10^{-6}$, the lowest level is slightly higher than at $q_B = 4.884 \times 10^{-7}$. When $q_B = 4.884 \times 10^{-5}$, only a few susceptible cells is infected. Subsequently, as the infection is controlled, the concentration of susceptible cells slowly increases. From Fig. 7b, we see that while the timing and rate of the initial rise in viral load are consistent, the viral loads decrease to zero earlier as q_B increases. This indicates that increasing the stimulation rate of B cells can lead to earlier control of viral infection in the lung, as free viruses are neutralized by the abundant antibodies.

We observe that the concentration of NK cells shows a slow increase initially, followed by a rapid rise at 3 days post-infection, and then a decrease due to infection control (see Fig. 7c). When $q_B = 4.884 \times 10^{-6}$, there is little change in the concen-

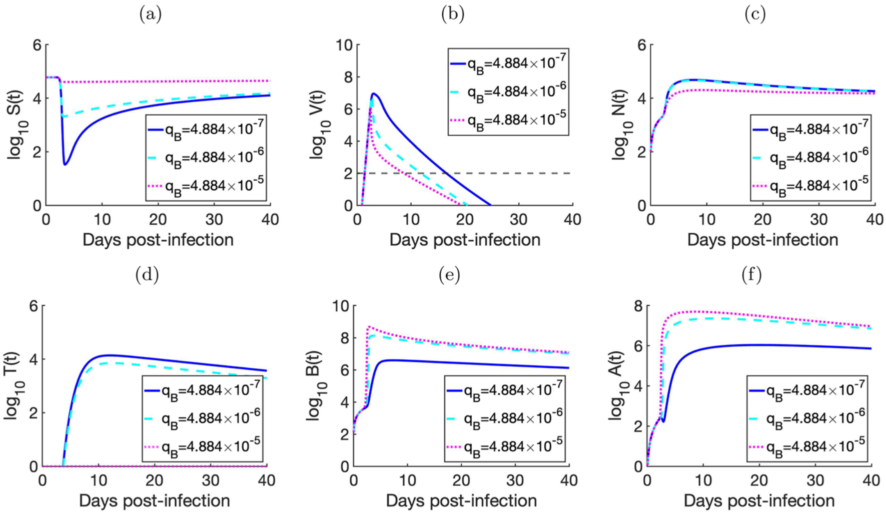


Fig. 7 The effect of increasing the stimulation rate of B cells q_B on the dynamics of the model (blue solid line for $q_B = 4.884 \times 10^{-7}$, cyan dashed line for $q_B = 4.884 \times 10^{-6}$, and purple dotted line for $q_B = 4.884 \times 10^{-5}$). The black dashed line in (**b**) represents the RT-PCR detection limit (Color figure online)

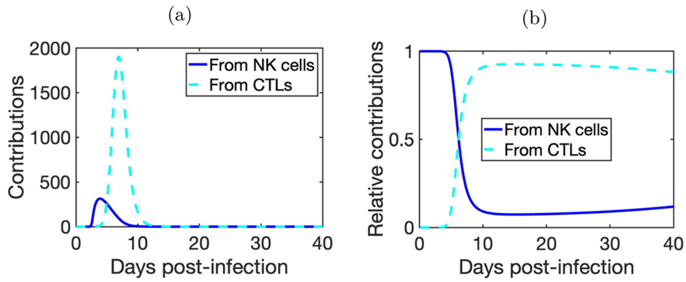


Fig. 8 The contributions to infected cells from NK cells and CTLs. **a** The blue solid line represents $\delta_N IN$, which is the rate of cytotoxic effect from NK cells; the cyan dashed line represents $\delta_T IT$, which is the rate of cytotoxic effect from CTLs. **b** The relative contributions to infected cells from NK cells and CTLs. The switch in relative contributions occurs around days 6 to 7 (Color figure online)

tration of NK cells compared to $q_B = 4.884 \times 10^{-7}$. However, at $q_B = 4.884 \times 10^{-5}$, the peak concentration of NK cells decreases significantly. For CTLs, increasing q_B to 4.884×10^{-6} does not change the activation time, but the peak value decreases slightly compared to $q_B = 4.884 \times 10^{-7}$ (see Fig. 7d). At $q_B = 4.884 \times 10^{-5}$, the concentration of CTLs is almost zero. This finding suggests that enhancing B cell stimulation may be more effective in clearing the virus from the lungs. Given that vaccines can effectively induce B cell responses and promote antibody production (Chen et al. 2022; Faiq 2020; Goel et al. 2021), vaccination is a viable strategy against SARS-CoV-2.

Although the model containing only cellular immunity (Model (A3)) cannot reduce the viral load below the RT-PCR detection limits, both NK cells and CTLs exhibit cytotoxic effects on infected cells in Model (1). We then evaluate the contributions of NK cells and CTLs to the elimination of infected cells, as shown in Fig. 8. From Fig. 8a, it can be observed that NK cells start to exert their effects between days 2 and 3 post-infection, peak on day 4, and gradually decline thereafter. The immune response induced by NK cells is generally rapid, often occurring within a few hours after infection. However, the simulation in this paper suggests that NK cells begin to function around the third day post-infection. We speculate that there may be delayed or reduced NK cell responses in COVID-19 patients. This result is consistent with that reported in Zhou et al. (2021), which indicates that the SARS-CoV-2 virus, in comparison to SARS-CoV and MERS-CoV, tends to elicit a delayed or diminished antiviral innate immune response during the early stages of infection. The CTL-mediated cellular immune response becomes activated on days 4 to 5, reaching its peak on days 7 to 8.

We also compare the relative contributions of these two cytotoxic effects on infected cells, as indicated by the following two ratios:

$$\frac{\delta_N IN}{\delta_N IN + \delta_T IT} \text{ and } \frac{\delta_T IT}{\delta_N IN + \delta_T IT},$$

as shown in Fig. 8b. During the early stage of infection, all cytotoxic effects on infected cells come from NK cells. Four days post-infection, the contribution from NK cells decreases, while the contribution from CTLs increases. The switch in relative contribution from the two cell types occurs around days 6 to 7 (see Fig. 8b). As the infection progresses, the killing effects from CTLs account for the majority (more than 90%). In summary, NK cells play a larger role in killing infected cells during the early stages of infection, but their contribution diminishes as CTLs take over. The results from Fig. 8a and b suggest that, in terms of cytotoxic effects, CTLs are stronger and more sustained than NK cells.

5 Complex Dynamics

In this section, we mainly examine the complex dynamics of System (1).

5.1 Stability Analysis

In this subsection, we investigate the existence and the local stability of equilibria for System (1). To enhance readability, we only present theorems here, and all proofs are provided in Appendix B.

Define

$$\begin{aligned} D = \{&(S, I, V, N, T, B, A) \in \mathbb{R}^7_+ \mid \\ &0 \leqslant S, I \leqslant \frac{\mu}{\min\{d_S, d_I\}}, 0 \leqslant V \leqslant \frac{p\mu}{c\min\{d_S, d_I\}}, \\ &0 \leqslant N \leqslant \frac{\tau \min\{d_S, d_I\} + q_N \mu}{d_N \min\{d_S, d_I\}}, \end{aligned}$$

$$0 \leqslant T \leqslant \frac{q_T \mu}{\delta_T \min\{d_S, \delta, d_T\}}, 0 \leqslant B \leqslant \hat{B}, 0 \leqslant A \leqslant \frac{f \hat{B}}{d_A}\},$$

where $\hat{B} = \frac{(q_B p\mu - d_B c \min\{d_S, d_I\}) + \sqrt{(q_B p\mu - d_B c \min\{d_S, d_I\})^2 + 4e\eta c^2 \min\{d_S, d_I\}^2}}{2ec \min\{d_S, d_I\}}$. Then we have

Theorem 1 *All solutions of model (1) remain nonnegative. Moreover, D is a global attractor in $\mathbb{R}^7_+$ and positively invariant for model (1).*

System (1) always has an infection-free equilibrium

$$E_0 = \left(\frac{\mu}{d_S}, 0, 0, \frac{\tau}{d_N}, 0, w, \frac{fw}{d_A}\right),$$

where $w = \frac{-d_B + \sqrt{d_B^2 + 4e\eta}}{2e}$. Define the matrices

$$\mathbf{F} = \begin{pmatrix} 0 & \frac{\beta\mu}{d_S} \\ 0 & 0 \end{pmatrix}, \mathbf{V} = \begin{pmatrix} \frac{\delta_N \tau}{d_N} + d_I & 0 \\ -p & \frac{kfw}{d_A} + c \end{pmatrix}.$$

By using the next-generation method (Van den Driessche and Watmough 2002), the basic reproduction number R_0 can be expressed as

$$R_0 = \rho(\mathbf{F}\mathbf{V}^{-1}) = \frac{\mu\beta p d_N d_A}{d_S(\delta_N \tau + d_I d_N)(kfw + cd_A)},$$

where $\rho(\mathbf{F}\mathbf{V}^{-1})$ represents the spectral radius of matrix $\mathbf{F}\mathbf{V}^{-1}$. The basic reproduction number R_0 represents the average number of new cell infections induced by the introduction of a single infected cell into a wholly susceptible target cell population (Guo et al. 2021).

Theorem 2 *If $R_0 < 1$, then the infection-free equilibrium E_0 is locally asymptotically stable; if $R_0 > 1$, then the infection-free equilibrium is unstable.*

Define

$$\Theta = \beta\mu d_N d_S \gamma p - [\mu d_N d_A p \beta^2 + d_S^2 \gamma c(\delta_N \tau + d_I d_N)],$$
$$\hat{V} = \frac{d_S[d_I d_T q_T d_N + \delta_N d_T(\tau q_T + d_T q_N)]}{\beta[\mu d_N q_T^2 - [d_I d_T q_T d_N + \delta_N d_T(\tau q_T + d_T q_N)]]}.$$

Theorem 3 *(a) Suppose that $R_0 > 1$. Then System (1) has at least one non-CTLs boundary equilibrium $E^\partial = (S^\partial, I^\partial, V^\partial, N^\partial, 0, B^\partial, A^\partial)$;*

(b) Suppose that $R_0 < 1$. Then the backward bifurcation may occur in the boundary $T = 0$ for System (1), and System (1) may have 0, 1, 2 non-CTLs boundary equilibria. Moreover, we have

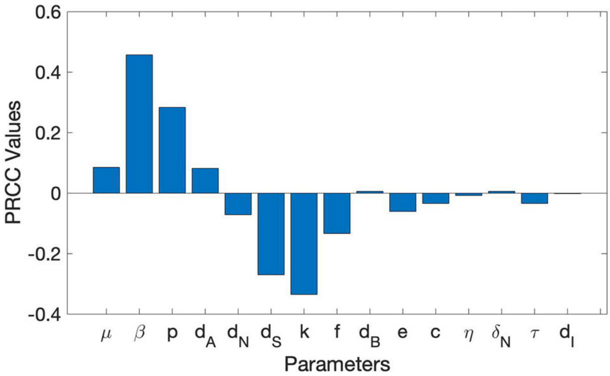


Fig. 9 Parameter sensitivity analysis for R_0 (Color figure online)

(i) if $\Theta > 0$*, System* (1) *may have* 0, 1, 2 *non-CTLs boundary equilibria;*
(ii) if $\Theta < 0$*, System* (1) *has no non-CTLs boundary equilibrium.*

Theorem 4 *(a) Suppose that* $\mu d_N q_T^2 < d_I d_T q_T d_N + \delta_N d_T(\tau q_T + d_T q_N)$. *Then System* (1) *has no positive equilibrium;*

(b) Suppose that $\mu d_N q_T^2 > d_I d_T q_T d_N + \delta_N d_T(\tau q_T + d_T q_N)$. *Then we have*

(i) if $\frac{(pd_T - cq_T\hat{V})(\gamma\hat{V} + d_A)}{fkq_T\hat{V}} > \frac{-(d_B - q_B\hat{V}) + \sqrt{(d_B - q_B\hat{V})^2 + 4e\eta}}{2e}$*, System* (1) *has a unique positive equilibrium* E^**. Moreover, the unique positive equilibrium* E^* *is either locally asymptotically stable, or the dimension of its unstable manifold and center manifold are both even;*
(ii) if $\frac{(pd_T - cq_T\hat{V})(\gamma\hat{V} + d_A)}{fkq_T\hat{V}} < \frac{-(d_B - q_B\hat{V}) + \sqrt{(d_B - q_B\hat{V})^2 + 4e\eta}}{2e}$*, System* (1) *has no positive equilibrium.*

5.2 Bifurcation Analysis and Simulation

In this subsection, due to the high dimension of System (1), we perform some bifurcation simulations to investigate the complex dynamics of System (1).

To determine which parameters have the greatest impact on the dynamic behaviors, a parameter sensitivity analysis for R_0 is performed using partial rank correlation coefficients ($PRCCs$), as shown in Fig. 9. From Fig. 9, we can see that the viral infection rate β has a significant impact on R_0, indicated by $|PRCC| > 0.4$.

Based on the sensitivity analysis for R_0, the viral infection rate β is selected as the bifurcation parameter to explore the complex dynamical behavior of System (1). The other parameter values are fixed as shown in Table 1, except for a slight variation in γ ($\gamma = 1.714 \times 10^{-7}$) to facilitate quicker convergence to equilibrium.

We first investigate the dynamic behavior of System (1) on the boundary. By setting $T = 0$, we obtain a 6-dimensional boundary system, which indicates that no effective CTLs immune response has been activated within the body. This is possible because in certain severe cases of COVID-19, the functionality of SARS-CoV-2-specific CTLs may be compromised, which correlates with the severity of the disease (Ogura et al. 2022). By using MatCont, we generate the bifurcation diagram of the boundary system, as shown in Fig. 10a. From Fig. 10a, it shows that as β increases, the 6-dimensional system undergoes one branch point and two Hopf bifurcation points at $\beta = \beta^{BP,1}$, $\beta = \beta^{H,1}$, and $\beta = \beta^{H,2}$, respectively.

If $\beta < \beta^{BP,1}$, then $R_0 < 1$, and according to Theorem 2, the infection-free equilibrium E_0 is locally asymptotically stable. If $\beta > \beta^{BP,1}$, then $R_0 > 1$, and the infection-free equilibrium E_0 becomes unstable. From the above discussion, it is clear that the boundary system of Model (1) undergoes a forward bifurcation at $\beta = \beta^{BP,1}$, leading to the emergence of the non-CTLs boundary equilibrium E_∂ in Model (1) when $\beta > \beta^{BP,1}$.

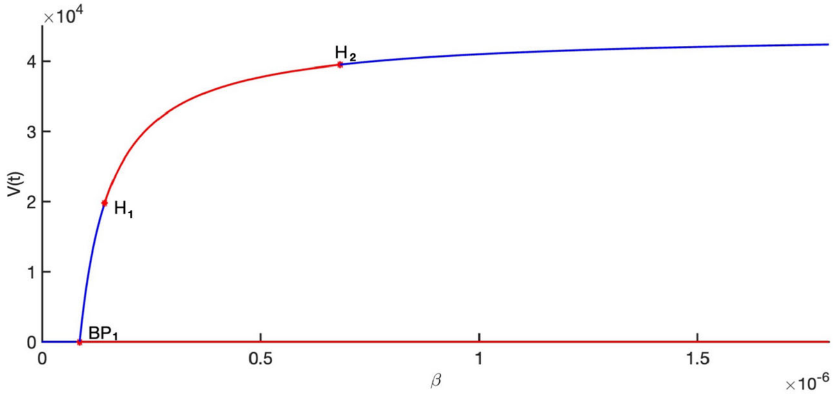
From Fig. 10a, we also observe that if $\beta^{BP,1} < \beta < \beta^{H,1}$ and $\beta > \beta^{H,2}$, the non-CTLs boundary equilibrium E_∂ is locally asymptotically stable on the boundary $T = 0$. However, if $\beta^{H,1} < \beta < \beta^{H,2}$, the non-CTLs boundary equilibrium E_∂ becomes unstable. Since the first Lyapunov coefficient of the 6-dimensional boundary system at $\beta = \beta^{H,1}$ is -2.106×10^{-4}, it follows that $\beta = \beta^{H,1}$ is a supercritical Hopf bifurcation point (Kuznetsov et al. 1998). This means that the non-CTLs boundary equilibrium E_∂ transitions from a stable state to an unstable state on the boundary $T = 0$, and the boundary system develops a stable periodic solution as β passes through the Hopf bifurcation point $\beta = \beta^{H,1}$ from the left.

Conversely, the first Lyapunov coefficient of the 6-dimensional boundary system at $\beta = \beta^{H,2}$ is 5.528×10^{-4}, indicating that $\beta = \beta^{H,2}$ is a subcritical Hopf bifurcation point (Kuznetsov et al. 1998). This implies that the non-CTLs boundary equilibrium E_∂ undergoes a transition from a stable state to an unstable state (Strogatz 2018), and the boundary system also develops an unstable periodic solution. Thus, in this case, the boundary system may have two periodic solutions: one stable and one unstable, which is confirmed by numerical simulations (see Fig. 12g).

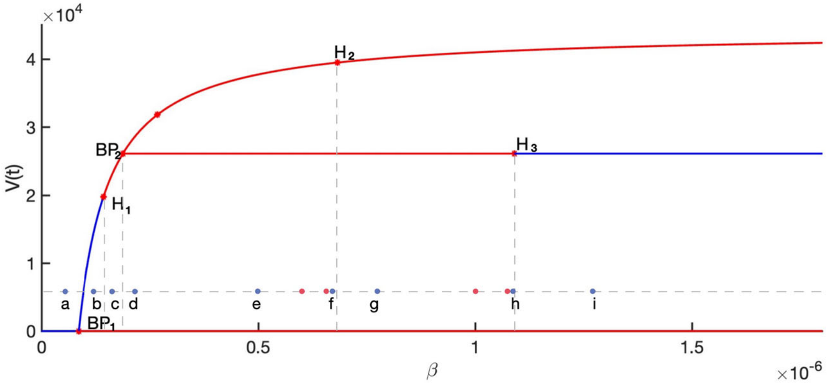
We then perform the bifurcation analysis for System (1), and the bifurcation diagram is shown in Fig. 10b. From Fig. 10b, it is clear that as β increases, System (1) undergoes a forward bifurcation at $\beta = \beta^{BP,1}$, generating a new non-CTLs boundary equilibrium E_∂. The non-CTLs boundary equilibrium E_∂ experiences one Hopf bifurcation point, one branch point, one neutral saddle equilibrium, and another Hopf bifurcation point at $\beta = \beta^{H,1}$, $\beta = \beta^{BP,2}$, $\beta = \beta^{NSE}$, and $\beta = \beta^{H,2}$, respectively, as β continues to increase.

The bifurcation dynamics of E_∂ on the boundary $T = 0$ has been discussed in the previous paragraph. When $\beta > \beta^{BP,2}$, the non-CTLs boundary equilibrium E_∂ is always unstable, and System (1) exhibits a positive equilibrium E^*. The positive equilibrium E^* undergoes a Hopf bifurcation at $\beta = \beta^{H,3}$. If $\beta^{BP,2} < \beta < \beta^{H,3}$, the positive equilibrium E^* is unstable; if $\beta > \beta^{H,3}$, the positive equilibrium E^* becomes locally asymptotically stable.

Since the first Lyapunov coefficient of System (1) at $\beta = \beta^{H,3}$ is -2.602×10^{-4}, it follows from Kuznetsov et al. (1998) that a supercritical Hopf bifurcation occurs at $\beta = \beta^{H,3}$ for System (1). In this case, the positive equilibrium undergoes a transition from a stable state to an unstable state, and System (1) exhibits a stable periodic solution as β passes through the Hopf bifurcation point $\beta = \beta^{H,3}$ from the right. The limit cycles near the Hopf bifurcation point $\beta^{H,3}$ are shown in Fig. 11.



(a)



(b)

Fig. 10 Bifurcation diagram. BP denotes Branch Point and H denotes Hopf bifurcation point. Blue lines represent stable equilibrium, while red lines represent unstable equilibrium. **a** Bifurcation diagram of System (1) on the boundary. The numerical simulation shows one Branch Point and two Hopf bifurcation points (H) at $\beta^{BP,1} = 8.596 \times 10^{-8}$, $\beta^{H,1} = 1.434 \times 10^{-7}$, and $\beta^{H,2} = 6.819 \times 10^{-7}$, respectively, as β increases. The first Lyapunov coefficients of the 6-dimensional system ($T = 0$) at $\beta = \beta^{H,1}$ and $\beta = \beta^{H,2}$ are -2.106×10^{-4} and 5.528×10^{-4}, respectively. **b** Bifurcation diagram of System (1). The numerical simulation shows one Branch Point, one Hopf bifurcation point (H), another Branch Point, and two additional Hopf bifurcations (H) at $\beta^{BP,1} = 8.596 \times 10^{-8}$, $\beta^{H,1} = 1.434 \times 10^{-7}$, $\beta^{BP,2} = 1.883 \times 10^{-7}$, $\beta^{H,2} = 6.819 \times 10^{-7}$, and $\beta^{H,3} = 1.090 \times 10^{-7}$, respectively, as β increases. The first Lyapunov coefficients of System (1) at $\beta = \beta^{H,1}$, $\beta = \beta^{H,2}$, and $\beta = \beta^{H,3}$ are -6.270×10^{-5}, 3.682×10^{-4}, and -2.602×10^{-4}, respectively (Color figure online)

To investigate the complex dynamics of System (1), various values of β near the bifurcation points are chosen to simulate the system's behavior. These behaviors are illustrated in Fig. 12. The parameter values used in Fig. 12 (a)$\sim$(i) are $\beta = 8.2 \times 10^{-8}$, 1.2×10^{-7}, 1.6×10^{-7}, 2.0×10^{-7}, 5.0×10^{-7}, 6.7×10^{-7}, 8.0×10^{-7}, 1.07×10^{-6}, 1.3×10^{-6}, respectively.

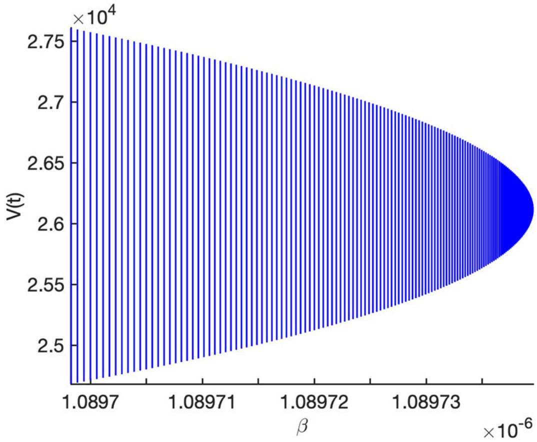


Fig. 11 Limit cycle simulations near the Hopf bifurcation point $\beta^{H,3}$ (Color figure online)

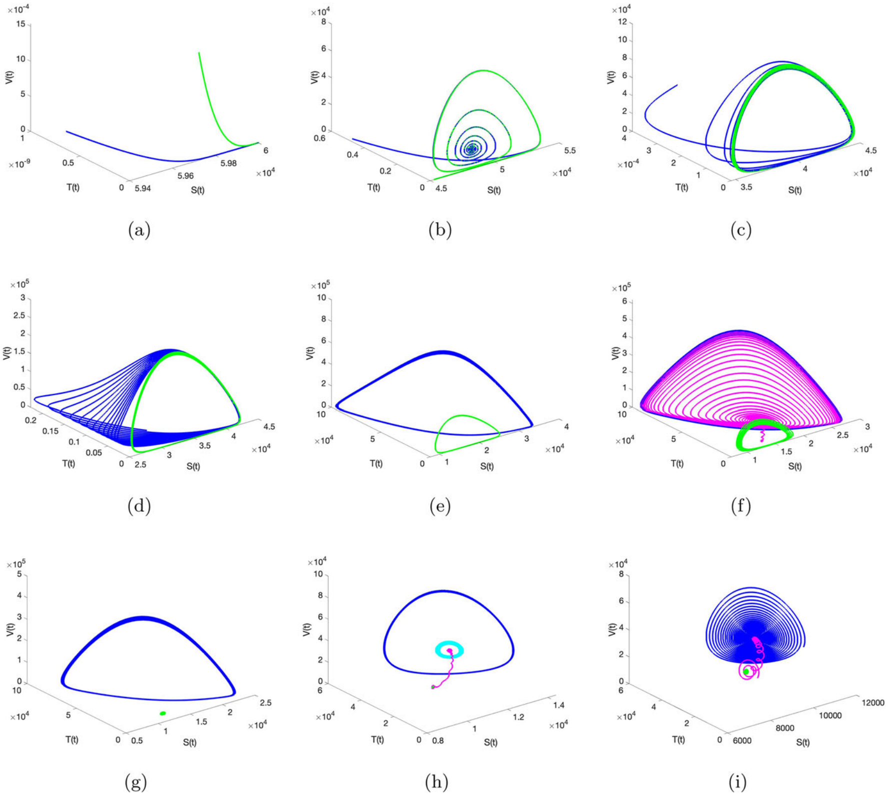


Fig. 12 Numerical simulations of System (1) for various β values chosen according to the blue point in the Hopf bifurcation diagram in Fig. 10: **a** $\beta = 8.2 \times 10^{-8}$, **b** $\beta = 1.2 \times 10^{-7}$, **c** $\beta = 1.6 \times 10^{-7}$, **d** $\beta = 2.0 \times 10^{-7}$, **e** $\beta = 5.0 \times 10^{-7}$, **f** $\beta = 6.7 \times 10^{-7}$, **g** $\beta = 8.0 \times 10^{-7}$, **h** $\beta = 1.07 \times 10^{-6}$, and **i** $\beta = 1.3 \times 10^{-6}$. The trajectory from the interior is marked in blue, and the trajectory from the boundary $T = 0$ is marked in green (Color figure online)

From Fig. 12a, where $\beta < \beta^{BP,1}$, all solutions converge to the infection-free equilibrium E_0. In Fig. 12b, System (1) shows a unique stable non-CTLs boundary equilibrium E_∂, with all solutions converging to the non-CTLs boundary equilibrium. In Fig. 12c, $\beta = 1.6 \times 10^{-7}$, which is greater than $\beta^{H,1}$. Since a Hopf bifurcation occurs at $\beta = \beta^{H,1}$, the system produces a stable boundary periodic solution, and the simulation results show that all solutions converge to this periodic solution. In Fig. 12d, where β passes through the Branch Point $\beta^{BP,2}$, System (1) exhibits two stable periodic solutions: one on the boundary $T = 0$ (green line) and the other in the interior (blue line).

In Fig. 12e, where $\beta = 5.0 \times 10^{-7}$, we observe that the system has a stable positive periodic solution and a stable boundary periodic solution. In Fig. 12f, where $\beta = 6.7 \times 10^{-7} < \beta^{H,2}$, the boundary periodic solution is unstable for System (1), and a slight perturbation of initial conditions causes the solutions to converge to the stable interior periodic solution. In Fig. 12g, where $\beta = 8.0 \times 10^{-7}$, System (1) has one stable non-CTLs boundary equilibrium point E_∂ and one stable positive periodic solution. In Fig. 12h, where $\beta = 1.07 \times 10^{-6} < \beta^{H,3}$, the boundary periodic solution is unstable for System (1), and a slight perturbation of initial conditions causes the solutions to converge to the stable interior equilibrium. We conjecture that in this case, the system has at least two positive periodic solutions. In Fig. 12i, where $\beta = 1.3 \times 10^{-6}$, which is greater than $\beta^{H,3}$, the positive equilibrium point E^* becomes stable, while the non-CTLs boundary equilibrium point E_∂ is unstable. The simulation results show that all solutions converge to the positive equilibrium E^*.

To further investigate the complex behavior of System (1), we also selected other values of β that are far from the bifurcation points. The chosen values of β are $\beta = 6.2 \times 10^{-7}, 6.6 \times 10^{-7}, 1.00 \times 10^{-6}, 1.06 \times 10^{-6}$, and the simulation results are shown in Fig. 13.

In Fig. 13a, where $\beta = 6.2 \times 10^{-7}$, we observe that System (1) has a stable positive periodic solution and a stable boundary periodic solution. When $\beta = 6.6 \times 10^{-7}$, System (1) exhibits more than three stable boundary periodic solutions and one stable positive periodic solution (see Fig. 13b). In Fig. 13c, where $\beta = 1.00 \times 10^{-6}$, System (1) has one stable non-CTLs boundary equilibrium and one stable positive periodic solution. When $\beta = 1.07 \times 10^{-6}$, System (1) shows more than three positive periodic solutions and one stable non-CTLs boundary equilibrium (see Fig. 13d).

In Sect. 5.1, we have made every effort to analyze the local behaviors of the model theoretically. However, we are still unable to provide a rigorous proof for its global behaviors. Therefore, we conduct mathematical simulations to investigate the complex dynamics that the model may exhibit in Sect. 5.2, including multiple periodic solutions and multiple attractors. These results may correlate with real disease outcomes: (i) If the system tends toward the infection-free equilibrium, this suggests that, under certain parameter values, the immune system successfully controls and eradicates the virus. (ii) If the system approaches a stable periodic solution, this may imply a dynamic interaction between the virus and the immune system, maintaining a form of equilibrium. (iii) If the system reaches a stable positive equilibrium, it indicates that the immune system fails to control the virus, resulting in a state of stable coexistence. Given the variability in COVID-19 patient outcomes, different parameter values can

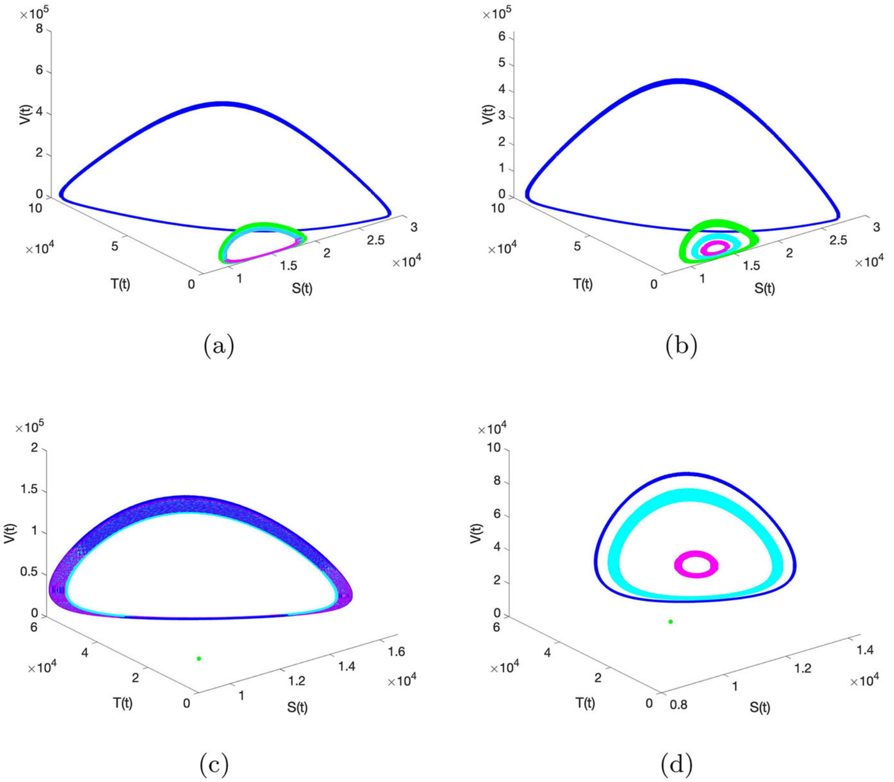


Fig. 13 Numerical simulations of System (1) for various β values chosen according to the red points in the Hopf bifurcation diagram in Fig. 10: $\beta = 6.2 \times 10^{-7}$, $\beta = 6.6 \times 10^{-7}$, $\beta = 1.00 \times 10^{-6}$, $\beta = 1.06 \times 10^{-6}$ (Color figure online)

lead to varying disease trajectories. Hence, collecting more clinical data is essential for further refinement of our model.

6 Conclusion and Discussion

In this paper, we formulated a within-host ODE model that describes the interaction between SARS-CoV-2 infection and the immune system. Our model includes the killing effect of NK cells and CTLs on infected cells, as well as the neutralization of free virus by antibodies secreted by B cells. The model effectively captures the acute short-term dynamics of the viral infection and, importantly, explores the synergy between NK cells, CTLs, and antibodies in combating SARS-CoV-2 infection. By evaluating the effects of different immune mechanisms, we gain some insights into the interaction between SARS-CoV-2 infection and the immune system from a mathematical perspective.

The approach used in our paper involves developing a mathematical model informed by biological mechanism and literature. Some parameters were obtained from existing studies, while more than half were estimated by fitting the model to the dataset. Once the parameters were determined, we conducted numerical simulations and analyses focused on specific biological aspects of interest. By comparing the fitting results of the five models with clinical data, we suggest that Model (1) best captures the viral dynamics of COVID-19 within the body. We explored the dynamics of NK cells, CTLs, B cells and antibodies utilizing the average values of the best-fitting parameters (Figs. 3, 4). We then investigated the synergy between NK cells, CTLs, and antibodies in viral clearance by comparing the viral load dynamics of System (1) with other SARS-CoV-2 infection models that incorporate one or two immune mechanisms (Fig. 5). Through setting the killing rate of NK cells $\delta_N = 0$, the killing rate of CTLs $\delta_T = 0$, or the neutralization rate by antibodies $k = 0$, we investigated the time series of viral load for Model (1), which indicate that antibodies play a crucial role compared to other immune mechanisms. We also illustrated the effect of increasing antibody production by elevating the stimulation rate of B cells on SARS-CoV-2 infection dynamics (Fig. 7), and the results show that enhancing B cell stimulation may be more effective in clearing the virus from the lungs; thus, vaccination is an effective strategy against SARS-CoV-2. Additionally, we evaluated the contributions of NK cells and CTLs to the elimination of infected cells (Fig. 8), which show that CTLs have stronger and longer-lasting cytotoxic effects compared to NK cells.

The existence and the local stability of equilibria of System (1) are investigated in Sect. 5.1. Due to the high dimensionality of System (1), we further explored its complex dynamic behavior through bifurcation analysis and numerical simulations (see Sect. 5.2). The numerical simulations of the boundary system show that it exhibits one forward bifurcation, one supercritical Hopf bifurcation, and one subcritical Hopf bifurcation (Fig. 10a). The numerical simulations of System (1) reveal that it undergoes two forward bifurcations, two supercritical Hopf bifurcations, and one subcritical Hopf bifurcation (Fig. 10b). The results also show that the model exhibits a rich variety of complex dynamics, including multiple periodic solutions (Fig. 13b, d) and multiple attractors (Fig. 12i). These outcomes, based on model's numerical simulations, may reflect the variability in disease progression observed clinically. However, whether these outcomes manifest in clinical practice remains unclear.

This study has a few limitations. The innate immune response involves various cells, such as NK cells, macrophages, neutrophils, and complement proteins (Tosi 2005). However, in this paper, we focus on the impact of NK cells on viral infection based on the following considerations. First, NK cells are a major type of lymphocytes, apart from T cells and B cells (Björkström et al. 2016). They originate from hematopoietic stem cells in the bone marrow and are considered core components of the innate immune system, accounting for 10–15% of the total lymphocyte count in human peripheral blood (Maskalenko et al. 2022). Second, they are found in the spleen, liver, lungs, bone marrow, and lymph nodes, and have the ability to spontaneously lyse infected cells. In our formulation of NK cells, we used a stimulation term that differs from those for CTLs and B cells, based on existing literature (Best et al. 2021). Unlike CTLs and B cells, which undergo antigen-driven clonal expansion, NK cells do not proliferate significantly in response to infection. Instead, they are primarily recruited and activated by cytokines produced by infected cells. This distinction justifies modeling NK cell production as dependent only on infected cells rather than their own population level. Various forms of proliferation terms exist in immune mechanisms (de Pillis et al. 2005; Adusei et al. 2015; Wang et al. 2013), and the influence of alternative formulations remains to be investigated.

To investigate the effect of the immune system on viral invasion, we selected RT-PCR data from eight mild or asymptomatic patients from Wölfel et al. (2020) for data fitting. Since COVID-19 is a highly heterogeneous disease (Karaderi et al. 2020; Ovsyannikova et al. 2020; Thomas et al. 2020), most symptomatic infected individuals can experience a wide range of symptoms, including fever, dry cough and muscle pain (Tay et al. 2020). Severe cases can develop acute respiratory distress syndrome (ARDS), characterized by difficulty in breathing and low blood oxygen level (Tay et al. 2020). Both viral infection and host responses play critical roles in determining disease severity and mortality in patients (Tay et al. 2020; Stebbing et al. 2020). Therefore, we will fit data from a larger population of patients in future work to obtain more generalizable results.

In our model and numerical simulation, we assumed that NK cell expansion occurs simultaneously with B cell expansion. This is because B cells are defined in this paper as undifferentiated B cells that have yet to differentiate into plasma cells. NK cells and B cells can be recruited to the infection site during the early stages of infection under the influence of chemokines (Sokol and Luster 2015). In reality, there might be a time delay between B cell activation and antibody production (Gujarati and Ambika 2014). To simplify the mathematical analysis, our model does not account for this time delay. Understanding how this potential delay impacts the dynamics warrants further study.

In summary, our model presents an advancement in describing the interaction between SARS-CoV-2 infection and the immune system, capturing the dynamic infection process from initial infection to the clearance, as well as the synergy of NK cells, CTLs, B cells, and antibodies in combating SARS-CoV-2 infection. Our study provides some insights into the complex interactions between viral infections and the immune system, offering valuable ideas for designing targeted therapies.

A Appendix

A.1 The Basic Virus Infection Model

The basic within-host model based on the target cell limited model is:

$$\begin{cases} \dfrac{\mathrm{d}S}{\mathrm{d}t} = \mu - d_S S - \beta V S, \\ \dfrac{\mathrm{d}I}{\mathrm{d}t} = \beta V S - d_I I, \\ \dfrac{\mathrm{d}V}{\mathrm{d}t} = pI - cV. \end{cases} \tag{A1}$$

The model keeps track of the numbers of susceptible cells (S), infected cells (I) and viral load (V) in the lung. In model (A1), the initial values of S, I, V are estimated to be 6×10^4 cells ml^{-1} (Wang et al. 2020), 0 cells ml^{-1}, 10^{-3} RNA copies ml^{-1}

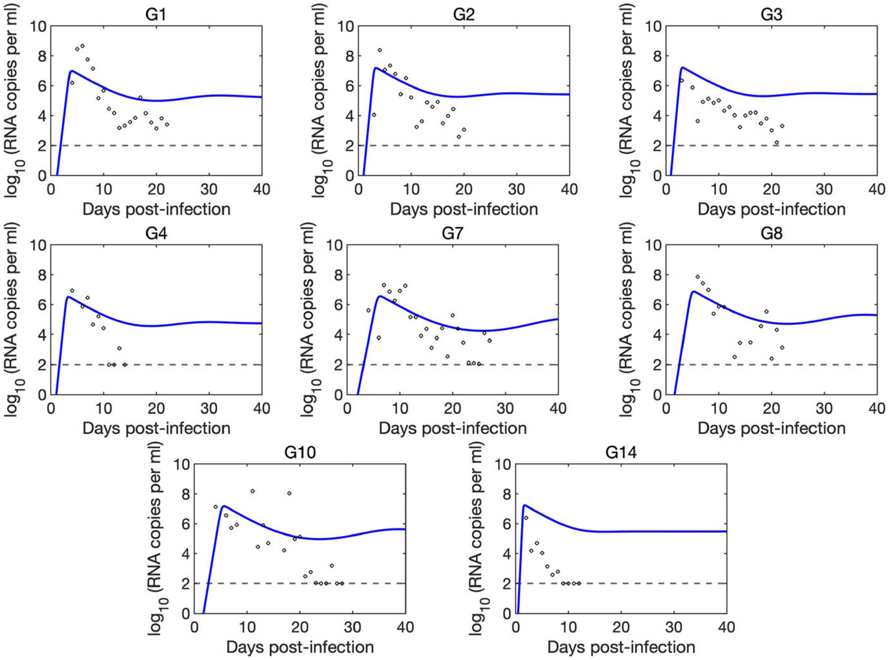


Fig. 14 Best-fitting results of each patients with (A1). The black circles are individual viral loads, and blue lines are the load projections of (A1) for each patients. The black dashed line represents the RT-PCR detection limit (Color figure online)

(Best et al. 2017; Wang et al. 2020; Ke et al. 2021), respectively. The base death rate of susceptible cells (d_S) is fixed to be 0.007 day^{-1} (Barkauskas et al. 2013; Desai et al. 2014); the death rate of infected cells (d_I) is assumed to be 0.47 day^{-1} (Dobrovolny et al. 2013), and the clearance rate of virus (c) is hypothesized to be 10 day^{-1} (Ke et al. 2021). The infection rate of the virus (β) and the production rate of the virus (p) are obtained by fitting with the clinical data from Munich (Wölfel et al. 2020). The best-fitting results are shown in Fig. 14. The best-fitting parameter values are shown in Table 4.

Table 4 Best-fitting parameter values of Model (A1) to the viral load data

Patient	β(ml/virus/day)	p(/day)
#1	9.713×10^{-7}	2101
#2	8.382×10^{-7}	3192
#3	8.725×10^{-7}	3332
#4	3.620×10^{-6}	673
#7	1.269×10^{-7}	853
#8	8.274×10^{-7}	1688
#10	3.927×10^{-7}	3357
#14	2.397×10^{-6}	3361
Mean	1.399×10^{-6}	2320
SD	1.073×10^{-6}	1149.9

A.2 The Model with NK Cells

The second model incorporating NK cells is formulated as follows:

$$\begin{cases} \dfrac{\mathrm{d}S}{\mathrm{d}t} = \mu - d_S S - \beta V S, \\ \dfrac{\mathrm{d}I}{\mathrm{d}t} = \beta V S - \delta_N I N - d_I I, \\ \dfrac{\mathrm{d}V}{\mathrm{d}t} = pI - cV, \\ \dfrac{\mathrm{d}N}{\mathrm{d}t} = \tau + q_N I - d_N N. \end{cases} \tag{A2}$$

The model keeps track of the concentration of NK cells in the lung, denoted as N, besides the variables in Model (A1). The initial number of NK cells $N(0)$ is assumed to be 100 cells ml^{-1}. The recruitment rate of NK cells (τ) is fixed to be 1×10^3 cells ml^{-1} day^{-1} (Wang et al. 2020; Zhang et al. 2007). The death rate of NK cells (d_N) is estimated to be 0.07 cells day^{-1} (Zhang et al. 2007). The killing rate on infected cells by NK cells (δ_N) and the activation rate of NK cells (q_N) are obtained by fitting with the clinical data from Munich (Wölfel et al. 2020). The best-fitting results are shown in Fig. 15. The best-fitting parameter values are shown in Table 5.

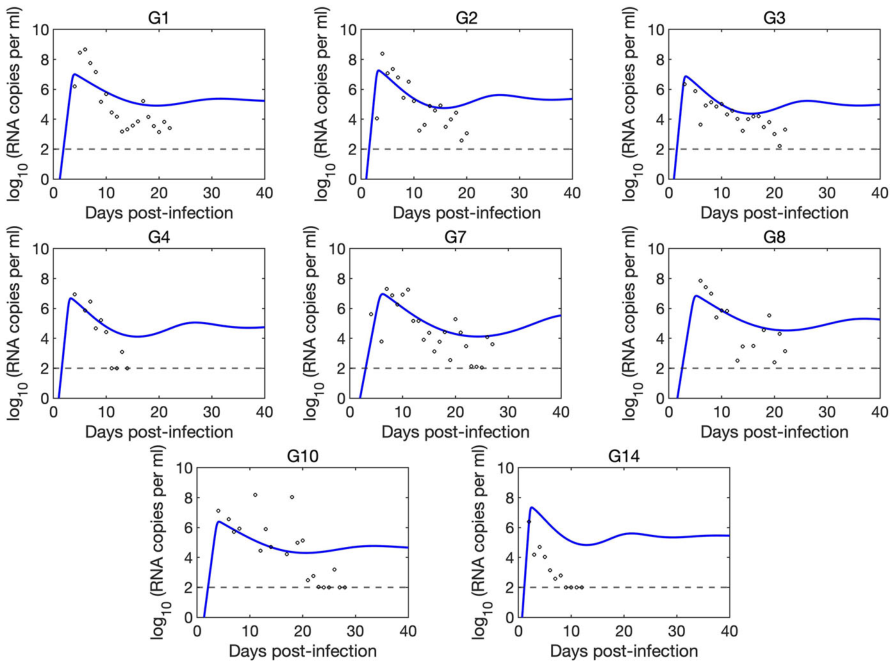


Fig. 15 Best-fitting results of each patients with (A2). The black circles are individual viral loads, and blue lines are the load projections of (A2) for each patients. The black dashed line represents the RT-PCR detection limit (Color figure online)

Table 5 Best-fitting parameter values of Model (A2) to the viral load data

Patient	β(ml/virus/day)	δ_N(/day)	p(/day)	q_N(/day)
#1	9.314×10^{-7}	6.743×10^{-6}	2134	0.602
#2	7.057×10^{-7}	6.249×10^{-6}	3823	0.600
#3	1.737×10^{-6}	6.019×10^{-6}	1547	0.600
#4	2.537×10^{-6}	6.314×10^{-6}	1000	0.600
#7	5.176×10^{-7}	3.088×10^{-6}	2203	0.599
#8	8.926×10^{-7}	9.432×10^{-7}	1570	0.600
#10	3.363×10^{-6}	7.397×10^{-7}	532.2	0.100
#14	9.197×10^{-7}	9.798×10^{-6}	4525	0.600
Mean	1.451×10^{-6}	4.987×10^{-6}	2167	0.538
SD	1.015×10^{-6}	3.132×10^{-6}	1367	0.177

Table 6 Best-fitting parameter values of Model (A3) to the viral load data

Patient	β(ml/virus/day)	δ_N(/day)	δ_T(/day)	p(/day)	q_N(/day)	q_T(/day)
#1	2.323×10^{-6}	6.004×10^{-8}	1.01×10^{-6}	970.8	0.100	1.580×10^{-4}
#2	6.323×10^{-7}	2.390×10^{-7}	4.415×10^{-6}	4831	0.447	1.668×10^{-4}
#3	3.322×10^{-6}	4.228×10^{-7}	1.020×10^{-5}	1073	0.447	1.582×10^{-4}
#4	2.535×10^{-6}	1.346×10^{-7}	1.147×10^{-5}	1687	0.199	1.750×10^{-4}
#7	5.314×10^{-7}	6.557×10^{-8}	5.312×10^{-5}	3261	0.116	1.315×10^{-4}
#8	8.926×10^{-7}	6.029×10^{-8}	7.000×10^{-5}	1688	0.100	1.190×10^{-4}
#10	9.726×10^{-7}	1.000×10^{-7}	5.320×10^{-5}	1522	0.100	1.252×10^{-4}
#14	3.697×10^{-6}	1.000×10^{-8}	9.320×10^{-6}	2000	0.100	1.800×10^{-4}
Mean	1.555×10^{-6}	1.664×10^{-7}	1.497×10^{-5}	3488	0.241	1.624×10^{-4}
SD	1.390×10^{-6}	1.243×10^{-7}	1.682×10^{-5}	1683	0.147	1.196×10^{-5}

A.3 The Model with NK Cells and CTLs

The third model involving both NK cells and cytotoxic T lymphocytes (CTLs) can be expressed as:

$$\begin{cases} \dfrac{\mathrm{d}S}{\mathrm{d}t} = \mu - d_S S - \beta V S, \\ \dfrac{\mathrm{d}I}{\mathrm{d}t} = \beta V S - \delta_N I N - \delta_T I T - d_I I, \\ \dfrac{\mathrm{d}V}{\mathrm{d}t} = pI - cV, \\ \dfrac{\mathrm{d}N}{\mathrm{d}t} = \tau + q_N I - d_N N, \\ \dfrac{\mathrm{d}T}{\mathrm{d}t} = q_T I T - d_T T. \end{cases} \tag{A3}$$

The model keeps track of the concentration of NK cells and CTLs in the lung, denoted as N and T, respectively, besides the variables in Model (A1). The initial value of CTLs $T(0)$ is fixed to be 0.001 cells ml^{-1} day^{-1}. As the half-life of CTLs ranges from several weeks to months, the natural death rate of CTLs is hypothesized to be 0.05 cells day^{-1}. The killing rate on infected cells by CTLs (δ_T) and the activation rate of CTLs (q_T) are obtained by fitting with the clinical data from Munich (Wölfel et al. 2020). The best-fitting results are shown in Fig. 16. The best-fitting parameter values are shown in Table 6.

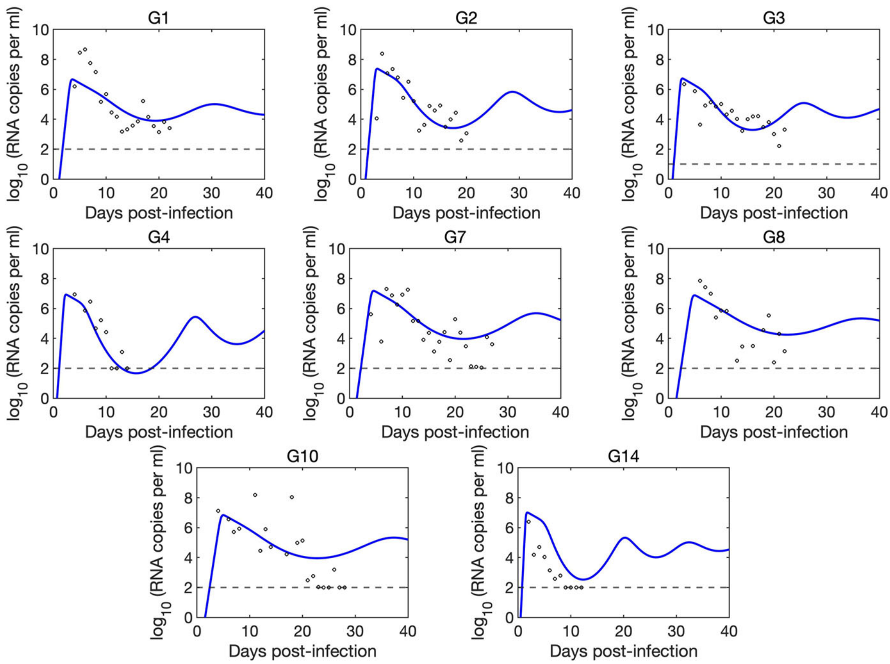


Fig. 16 Best-fitting results of each patients with (A3). The black circles are individual viral loads, and blue lines are the load projections of (A3) for each patients. The black dashed line represents the RT-PCR detection limit (Color figure online)

A.4 The Model with NK Cells, B Cells, and Antibodies

The fourth model involving both NK cells, B cells, and antibodies can be expressed as:

$$\begin{cases} \dfrac{\mathrm{d}S}{\mathrm{d}t} = \mu - d_S S - \beta V S, \\ \dfrac{\mathrm{d}I}{\mathrm{d}t} = \beta V S - \delta_N I N - d_I I, \\ \dfrac{\mathrm{d}V}{\mathrm{d}t} = pI - cV - kVA, \\ \dfrac{\mathrm{d}N}{\mathrm{d}t} = \tau + q_N I - d_N N, \\ \dfrac{\mathrm{d}B}{\mathrm{d}t} = \eta + q_B B V - d_B B - e B^2, \\ \dfrac{\mathrm{d}A}{\mathrm{d}t} = f B - \gamma A V - d_A A. \end{cases} \tag{A4}$$

The model keeps track of the concentration of NK cells, B cells, and antibodies in the lung, denoted as N, B, and A, respectively, besides the variables in Model (A1). The initial value of B cells $B(0)$ is fixed to be 130 cells ml^{-1} day^{-1}. The recruitment rate of B cells (η) is fixed to be 2×10^3 cells ml^{-1} day^{-1} (Wang et al. 2020; Cooper

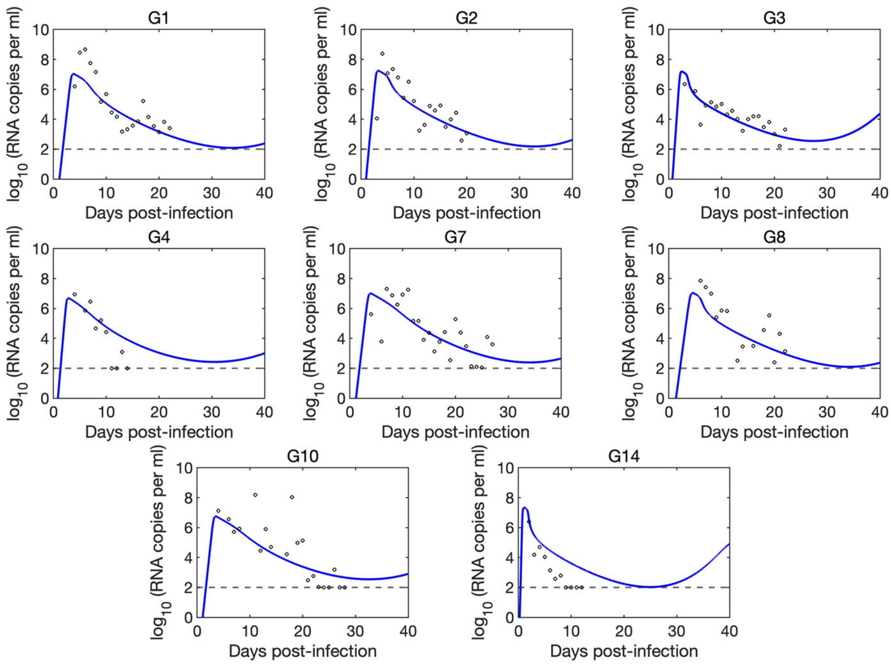


Fig. 17 Best-fitting results of each patients with (A4). The black circles are individual viral loads, and blue lines are the load projections of (A4) for each patients. The black dashed line represents the RT-PCR detection limit (Color figure online)

and Alder 2006). The death rate of B cells (d_B) is fixed at 0.033 cells day^{-1}, as the lifespan of B cells is about 4.3 weeks (Fulcher and Basten 1997). The initial number of antibodies $A(0)$ is fixed at 0 ng ml^{-1}. The half-life of antibodies in the body depends on the specific type of antibody, typically ranging from five days to one month (Barnes et al. 2021; Kinyanjui et al. 2007). Here, the antibody decay rate is assumed to be 0.12 day^{-1}. The stimulate rate of B cells (q_B), the density-dependent term of B cells (e), the production rate of antibodies by B cells (f), the loss rate of free antibodies in the formation of antibody-virus complexes (γ), and the neutralization rate by antibodies (k) are obtained by fitting with the clinical data from Munich (Wölfel et al. 2020). The best-fitting results are shown in Fig. 17. The best-fitting parameter values are shown in Table 7.

B Appendix B

Proof of Theorem 1. For model (1), we have $\frac{\mathrm{d}S}{\mathrm{d}t}|_{S=0} = \mu \geqslant 0$, $\frac{\mathrm{d}I}{\mathrm{d}t}|_{I=0} = \beta V S \geqslant 0$, $\frac{\mathrm{d}V}{\mathrm{d}t}|_{V=0} = pI \geqslant 0$, $\frac{\mathrm{d}N}{\mathrm{d}t}|_{N=0} = \tau \geqslant 0$, $\frac{\mathrm{d}T}{\mathrm{d}t}|_{T=0} = 0$, $\frac{\mathrm{d}B}{\mathrm{d}t}|_{B=0} = \eta \geqslant 0$, $\frac{\mathrm{d}A}{\mathrm{d}t}|_{A=0} = fB \geqslant 0$. This means that all trajectories of model (1) starting from $\mathbb{R}^7_+$ will enter and stay in this region $\mathbb{R}^7_+ = \{(S, I, V, N, T, B, A)|S(0) \geqslant 0, I(0) \geqslant 0, V(0) \geqslant 0, N(0) \geqslant 0, T(0) \geqslant 0, B(0) \geqslant 0, A(0) \geqslant 0\}$. From the first and second equations

Table 7 Best-fitting parameter values of Model (A4) to the viral load data

Patient	β(ml/virus/day)	δ_N(/day)	p(/day)	k(copies/day)	q_N(/day)
#1	9.731×10^{-7}	1.023×10^{-7}	2270	1.281×10^{-5}	0.444
#2	7.902×10^{-7}	2.561×10^{-8}	3615	1.393×10^{-5}	0.363
#3	1.416×10^{-6}	6.026×10^{-8}	3100	1.340×10^{-5}	0.106
#4	3.226×10^{-6}	6.386×10^{-7}	1005	7.981×10^{-5}	0.599
#7	9.736×10^{-7}	1.000×10^{-7}	2162	4.283×10^{-5}	0.420
#8	7.310×10^{-7}	4.928×10^{-8}	2372	2.040×10^{-6}	0.010
#10	1.999×10^{-6}	6.092×10^{-8}	1183	8.040×10^{-6}	0.010
#14	5.161×10^{-6}	9.308×10^{-7}	4244	1.721×10^{-5}	0.414
Mean	1.909×10^{-6}	2.460×10^{-7}	2493.9	2.376×10^{-5}	0.296
SD	1.556×10^{-6}	3.425×10^{-7}	1121.1	2.561×10^{-5}	0.223

Patient	q_B(cells/day)	e(ml/cells/day)	f(/day)	γ(/day)
#1	4.506×10^{-7}	9.300×10^{-10}	0.010	5.000×10^{-7}
#2	3.816×10^{-7}	8.962×10^{-10}	0.010	1.595×10^{-6}
#3	6.274×10^{-7}	9.112×10^{-10}	0.010	3.670×10^{-7}
#4	3.949×10^{-7}	1.016×10^{-8}	0.300	8.334×10^{-7}
#7	3.800×10^{-7}	1.800×10^{-9}	0.006	3.641×10^{-6}
#8	6.800×10^{-7}	2.970×10^{-10}	0.011	4.210×10^{-7}
#10	6.800×10^{-7}	1.200×10^{-10}	0.010	9.000×10^{-7}
#14	5.949×10^{-7}	9.441×10^{-10}	0.048	3.971×10^{-7}
Mean	5.237×10^{-7}	2.007×10^{-9}	0.051	1.082×10^{-6}
SD	1.349×10^{-7}	3.332×10^{-9}	0.102	1.112×10^{-6}

of (1), we have $\frac{\mathrm{d}(S+I)}{\mathrm{d}t} \leqslant \mu - d_S S - d_I I \leqslant \mu - \min\{d_S, d_I\}(S+I)$. It follows from Theorem 3.1 in Siewe and Friedman (2023) that $S(t) + I(t) \leqslant \frac{\mu}{\min\{d_S,d_I\}} + (S(0) + I(0) - \frac{\mu}{\min\{d_S,d_I\}})e^{-\min\{d_S,d_I\}t}$. Thus, we have $\limsup_{t\to\infty} \max\{S(t), I(t)\} \leqslant \frac{\mu}{\min\{d_S,d_I\}}$.

From the third and fourth equations of (1), we have $\frac{\mathrm{d}V}{\mathrm{d}t} \leqslant pI - cV \leqslant \frac{p\mu}{\min\{d_S,d_I\}} - cV$, and $\frac{\mathrm{d}N}{\mathrm{d}t} \leqslant \frac{q_N\mu}{\min\{d_S,d_I\}} - d_N N$. Using the same method as above, we obtain $\limsup_{t\to\infty} V(t) \leqslant \frac{p\mu}{c\min\{d_S,d_I\}}$ and $\limsup_{t\to\infty} N(t) \leqslant \frac{\tau\min\{d_S,d_I\}+q_N\mu}{d_N\min\{d_S,d_I\}}$. Define

$$H(t) = S(t) + I(t) + \frac{\delta_T}{q_T}T(t).$$

It then follows that

$$H'(t) = S'(t) + I'(t) + \frac{\delta_T}{q_T}T'(t) \leqslant \mu - \min\{d_S, d_I, d_T\}[S(t) + I(t) + \frac{\delta_T}{q_T}T(t)],$$

which implies $H(t) \leqslant \frac{\mu}{\min\{d_S,d_I,d_T\}} + [S(0) + I(0) + \frac{\delta_T}{q_T}T(0) - \frac{\mu}{\min\{d_S,d_I,d_T\}}]$ $e^{-\min\{d_S,d_I,d_T\}t}$. Taking the superior limit on both sides yields $\limsup_{t\to\infty} T(t) \leqslant \frac{q_T\mu}{\delta_T \min\{d_S,d_I,d_T\}}$.

From the sixth equation of System (1), we have $\frac{\mathrm{d}B}{\mathrm{d}t} \leqslant \eta + \frac{q_B p\mu - d_B c\min\{d_S,d_I\}}{c\min\{d_S,d_I\}}B - eB^2$. Direct calculation yields that $\limsup_{t\to\infty} B(t) \leqslant \hat{B}$. From the seventh equation of (1), we have $\frac{\mathrm{d}A}{\mathrm{d}t} \leqslant f\hat{B} - d_A A$. Thus, we obtain $\limsup_{t\to\infty} A(t) \leqslant \frac{f\hat{B}}{d_A}$. This completes the proof. □

Proof of Theorem 2. The variational matrix of the linearized system at E_0 is

$$\mathcal{J}(\mathcal{E}_I) = \begin{pmatrix} -d_S & 0 & -\frac{\beta\mu}{d_S} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\delta_N\tau}{d_N} - d_I & \frac{\beta\mu}{d_S} & 0 & 0 & 0 & 0 \\ 0 & p & -c - \frac{kfw}{d_A} & 0 & 0 & 0 & 0 \\ 0 & q_N & 0 & -d_N & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -d_T & 0 & 0 \\ 0 & 0 & q_B w & 0 & 0 & -d_B - 2ew & 0 \\ 0 & 0 & -\frac{\gamma fw}{d_A} & 0 & 0 & f & -d_A \end{pmatrix}.$$

The characteristic equation corresponding to this variational matrix is

$$\begin{aligned} G(\lambda) &= (\lambda + d_S)(\lambda + d_A)(\lambda + d_B + 2ew)(\lambda + d_T)(\lambda + d_N) \\ &\quad ((\lambda + \frac{\delta_N\tau}{d_N} + d_I)(\lambda + c + \frac{kfw}{d_A}) - \frac{\beta\mu p}{d_S}) = 0. \end{aligned}$$

Solving this equation, we get $\lambda_1 = -d_S$, $\lambda_2 = -d_A$, $\lambda_3 = -d_B - 2ew$, $\lambda_4 = -d_T$, $\lambda_5 = -d_N$, and

$$\lambda_{6,7} = \frac{-b \pm \sqrt{b^2 + 4\frac{(\delta_N\tau + d_I d_N)(cd_A + kfw)}{d_N d_A}(R_0 - 1)}}{2},$$

where $b = \frac{\delta_N\tau}{d_N} + d_I + c + \frac{kfw}{d_A}$. It can be seen that all eigenvalues have negative real parts if $R_0 < 1$. Thus, the infection-free equilibrium E_0 is locally asymptotically stable. If $R_0 > 1$, E_0 becomes unstable because $\lambda_6 > 0$. □

Proof of Theorem 3. For simplicity, we define

$$
\begin{aligned}
\hat{I} &= \frac{\sqrt{(\delta_N\tau + d_I d_N)^2 + 4\delta_N q_N \mu d_N} - (\delta_N\tau + d_I d_N)}{2\delta_N q_N},\\
\mathcal{F}(I) &= \frac{d_S\delta_N q_N I + d_S\delta_N\tau + d_S d_I d_N}{\beta[\mu d_N - (\delta_N\tau + d_I d_N)I - \delta_N q_N I^2]},\\
\hat{\omega}(I) &= \frac{\sqrt{(d_B - q_B I\mathcal{F}(I))^2 + 4e\eta} - (d_B - q_B I\mathcal{F}(I))}{2e},\\
\hat{R}(I) &= \frac{\mu\beta p d_N d_A}{d_S(\delta_N\tau + d_I d_N)(kf\hat{\omega}(I) + cd_A)},\\
\mathcal{G}(I) &= d_S(kf\hat{\omega}(I) + cd_A)[\delta_N q_N I + (\delta_N\tau + d_I d_N)](1 - \hat{R}(I)),\\
\mathcal{H}(I) &= I(d_S\gamma p - d_A p\beta - d_S\gamma c\mathcal{F}(I))[\delta_N q_N I + (\delta_N\tau + d_I d_N)].
\end{aligned}
$$

Any boundary equilibrium satisfies the following algebraic equations

$$
\begin{cases}
\mu - d_S S - \beta V S = 0,\\
\beta V S - \delta_N I N - d_I I = 0,\\
pI - cV - kVA = 0,\\
\tau + q_N I - d_N N = 0,\\
\eta + q_B B V - d_B B - eB^2 = 0,\\
fB - \gamma A V - d_A A = 0.
\end{cases}
\tag{B1}
$$

From the first and fourth equations of (B1), we have $\beta V S = \mu - d_S S$, and $N = \frac{\tau + q_N I}{d_N}$. Substituting the above equations into the second equation of (B1) yields that

$$
S = \frac{\mu d_N - (\delta_N\tau + d_I d_N)I - \delta_N q_N I^2}{d_S d_N}.
$$

Furthermore, we have $S > 0$ if and only if $0 < I < \hat{I}$. Since $\beta V S = \mu - d_S S$, it follows that

$$
V = I\mathcal{F}(I).
$$

Straightforward calculation yields that

$$
\begin{aligned}
\mathcal{F}'(I) = &\frac{d_S[\delta_N q_N \mu d_N + \delta_N^2 q_N^2 I^2 + (\delta_N\tau + d_I d_N)^2 + 2\delta_N q_N(\delta_N\tau + d_I d_N)]I}{\beta[\mu d_N - (\delta_N\tau + d_I d_N)I - \delta_N q_N I^2]^2}\\
&> 0,\ I \in (0, \hat{I}).
\end{aligned}
$$

Thus, $\mathcal{F}(I)$ is an increasing function of I in the interval $(0, \hat{I})$ and $\lim_{I\to\hat{I}} \mathcal{F}(I) = +\infty$. Moreover, V increases with I in the interval $(0, \hat{I})$.

From the fifth equation of (B1), we have

$$B = \frac{\sqrt{(d_B - q_B V)^2 + 4e\eta} - (d_B - q_B V)}{2e}.$$

Since

$$B'(V) = \frac{q_B\sqrt{(d_B - q_B V)^2 + 4e\eta} - q_B(d_B - q_B V)}{2e\sqrt{(d_B - q_B V)^2 + 4e\eta}} > 0,$$

it follows that B increases with V. Substituting $V = I\mathcal{F}(I)$ yields that

$$B = \hat{\omega}(I).$$

Because B increases with V and V increases with I in the interval $(0, \hat{I})$, $\hat{\omega}(I)$ is an increasing function of I in the interval $(0, \hat{I})$. It follows from the sixth equation of (B1) that $A = \frac{fB}{\gamma V + d_A}$. Since $V = I\mathcal{F}(I)$, we have

$$A = \frac{fB}{\gamma I\mathcal{F}(I) + d_A}.$$

By substituting $B = \hat{\omega}(I)$ and $A = \frac{fB}{\gamma I\mathcal{F}(I)+d_A}$ into the third equation of (B1), and after extensive calculations, we can obtain the following equation

$$\mathcal{G}(I) = \mathcal{H}(I). \tag{B2}$$

It follows from that the boundary equilibrium, if it exists, must satisfy the algebraic equation (B2). In the following, we will investigate the solutions of the equation (B2) in the interval $(0, \hat{I})$.

By direct calculation, we can conclude that the function $\mathcal{G}(I)$ satisfies: (i) $\mathcal{G}(0) = d_S(kf\omega + cd_A)(\delta_N\tau + d_I d_N)(1 - R_0)$; (ii) $\mathcal{G}(I)$ is an increasing function of I in the interval $(0, \hat{I})$; (iii) $\lim\limits_{I\to\hat{I}} \mathcal{G}(I) = +\infty$, and the function $\mathcal{H}(I)$ satisfies: (i)$\mathcal{H}(0) = 0$; (ii)$\mathcal{H}'(I)$ is monotonically decreasing with $\mathcal{H}'(0) = \frac{(\delta_N\tau+d_I d_N)}{\beta\mu d_N}\Theta$ and $\lim\limits_{I\to\hat{I}} \mathcal{H}'(I) = -\infty$; (iii) $\lim\limits_{I\to\hat{I}} \mathcal{H}(I) = -\infty$. Then we consider four cases.

Case 1. $R_0 > 1$, $\Theta > 0$. In this case, we have: (a) $\mathcal{G}(0) < 0$, $\lim\limits_{I\to\hat{I}} \mathcal{G}(I) = +\infty$ and $\mathcal{G}(I)$ is increasing in the interval $(0, \hat{I})$; (b) $\mathcal{H}(0) = 0$, $\lim\limits_{I\to\hat{I}} \mathcal{H}(I) = -\infty$ and $\mathcal{H}(I)$ is a concave function in the interval $(0, \hat{I})$. It follows that the equation (B2) has at least one solution (see Fig. 18a), denoted by $I = I^{\partial}$. Thus, System (1) has at least one

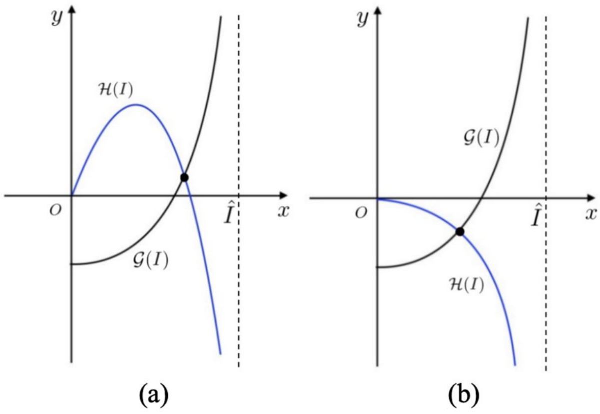


Fig. 18 Case $R_0 > 1$. **a** $\Theta > 0$: the equation (B2) has at least one solution; **b** $\Theta < 0$: the equation (B2) has a unique solution (Color figure online)

non-CTLs boundary equilibrium $E^\partial = (S^\partial, I^\partial, V^\partial, N^\partial, 0, B^\partial, A^\partial)$, where

$$S^\partial = \frac{\mu d_N - (\delta_N \tau + d_I d_N) I^\partial - \delta_N q_N {I^\partial}^2}{d_S d_N},$$
$$V^\partial = \frac{I^\partial (d_S \delta_N q_N I^\partial + d_S \delta_N \tau + d_S d_I d_N)}{\beta[\mu d_N - (\delta_N \tau + d_I d_N) I^\partial - \delta_N q_N {I^\partial}^2]},$$
$$N^\partial = \frac{\tau + q_N I^\partial}{d_N}, \; B^\partial = \frac{\sqrt{(d_B - q_B I^\partial \mathcal{F}(I^\partial))^2 + 4e\eta} - (d_B - q_B I^\partial \mathcal{F}(I^\partial))}{2e},$$
$$A^\partial = \frac{f B^\partial}{\gamma I^\partial \mathcal{F}(I^\partial) + d_A}.$$

Case 2. $R_0 > 1$, $\Theta < 0$. In this case, we have: (a) $\mathcal{G}(0) < 0$, $\lim_{I \to \hat{I}} \mathcal{G}(I) = +\infty$ and $\mathcal{G}(I)$ is increasing in the interval $(0, \hat{I})$; (b) $\mathcal{H}(0) = 0$, $\lim_{I \to \hat{I}} \mathcal{H}(I) = -\infty$ and $\mathcal{H}(I)$ is a monotonically decreasing and concave function in the interval $(0, \hat{I})$. It follows that the equation (B2) has a unique solution (see Fig. 18b). Thus, in this case System (1) has a unique non-CTLs boundary equilibrium $E^\partial = (S^\partial, I^\partial, V^\partial, N^\partial, 0, B^\partial, A^\partial)$.

Case 3. $R_0 < 1$, $\Theta > 0$. In this case, we have: (a) $\mathcal{G}(0) > 0$, $\lim_{I \to \hat{I}} \mathcal{G}(I) = +\infty$ and $\mathcal{G}(I)$ is increasing in the interval $(0, \hat{I})$; (b) $\mathcal{H}(0) = 0$, $\lim_{I \to \hat{I}} \mathcal{H}(I) = -\infty$ and $\mathcal{H}(I)$ is a concave function in the interval $(0, \hat{I})$. It follows that the equation (B2) has either 0, 1, or 2 solutions (see Fig. 19a, b, and c). Thus, in this case, System (1) may have 0, 1, 2 non-CTLs boundary equilibria.

Case 4. $R_0 < 1$, $\Theta < 0$. In this case, we have: (a) $\mathcal{G}(0) > 0$, $\lim_{I \to \hat{I}} \mathcal{G}(I) = +\infty$ and $\mathcal{G}(I)$ is increasing in the interval $(0, \hat{I})$; (b) $\mathcal{H}(0) = 0$, $\lim_{I \to \hat{I}} \mathcal{H}(I) = -\infty$ and $\mathcal{H}(I)$ is

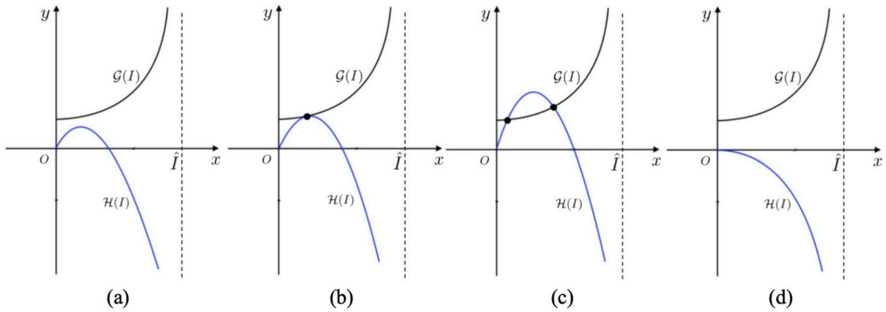


Fig. 19 Case $R_0 < 1$. **a** $\Theta > 0$: the equation (B2) has no solution; **b** $\Theta > 0$: the equation (B2) has one solution; **c** $\Theta > 0$: the equation (B2) has at least two solutions; **d** $\Theta < 0$: the equation (B2) has no solution (Color figure online)

a monotonically decreasing and concave function in the interval $(0, \hat{I})$. It follows that the equation (B2) has no solution (see Fig. 19d). Thus, in this case System (1) has no non-CTLs boundary equilibria. □

Proof of Theorem 4. For ease of notation, define

$$F(V) = \frac{(pd_T - cq_T V)(\gamma V + d_A)}{fkq_T V},$$
$$G(V) = \frac{-(d_B - q_B V) + \sqrt{(d_B - q_B V)^2 + 4e\eta}}{2e}.$$

Any positive equilibrium satisfies the following allegoric equations

$$\begin{cases} \mu - d_S S - \beta V S = 0, \\ \beta V S - \delta_N I N - \delta_T I T - d_I I = 0, \\ pI - cV - kVA = 0, \\ \tau + q_N I - d_N N = 0, \\ q_T I - d_T = 0, \\ \eta + q_B B V - d_B B - eB^2 = 0, \\ fB - \gamma A V - d_A A = 0. \end{cases} \tag{B3}$$

From the fifth and seventh equations of (B3), we have $I = \frac{d_T}{q_T}$ and $A = \frac{fB}{\gamma V + d_A}$. Substituting the above equations into the third equation of (B3) yields that

$$B = F(V). \tag{B4}$$

Furthermore, we have $B > 0$ if and only if $0 < V < \frac{pd_T}{cq_T}$. From the sixth equation of (B3), we have

$$B = G(V). \tag{B5}$$

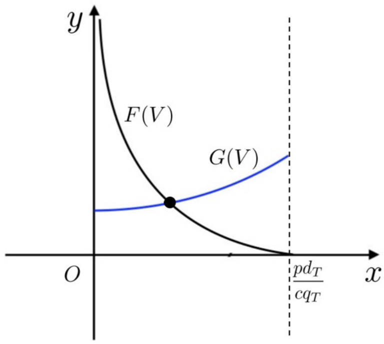


Fig. 20 The equation $F(V) = G(V)$ has a unique solution in the interval $(0, \frac{pd_T}{cq_T})$ (Color figure online)

It follows from (B4) and (B5) that the positive equilibrium, if it exists, must satisfy the algebraic equation $F(V) = G(V)$. In the following, we will investigate the solutions of the equation $F(V) = G(V)$ in the interval $(0, \frac{pd_T}{cq_T})$.

Direct calculation yields that $F'(V) < 0$ for all V. Thus, $F(V)$ is a decreasing function of V in the interval $(0, \frac{pd_T}{cq_T})$. Moreover, $\lim\limits_{V\to 0^+} F(V) = +\infty$ and $F\left(\frac{pd_T}{cq_T}\right) = 0$. Since $G'(V) > 0$ for all V, it follows that $G(V)$ increases with V. Moreover, $\mathcal{G}(0) = \omega > 0$, where ω is define in Sect. 5.1. This means that equation $F(V) = G(V)$ has a unique solution in the interval $(0, \frac{pd_T}{cq_T})$ (see Fig. 20). We denote the solution as V^*.

From the first equation of (B3) yields that $S = \frac{\mu}{d_S+\beta V^*}$. Substituting V^*, $S = \frac{\mu}{d_S+\beta V^*}$, and $I = \frac{d_T}{q_T}$ into the second equation of (B3) yields that

$$T = \frac{q_T}{\delta_T d_T}\Big[\frac{\mu\beta V^*}{d_S + \beta V^*} - \frac{d_I d_T q_T d_N + \delta_N d_T(\tau q_T + d_T q_N)}{d_N q_T^2}\Big].$$

Then we have:

(a) Suppose that $\mu d_N q_T^2 < d_I d_T q_T d_N + \delta_N d_T(\tau q_T + d_T q_N)$. Then $\frac{\mu\beta V^*}{d_S+\beta V^*} < \frac{d_I d_T q_T d_N+\delta_N d_T(\tau q_T+d_T q_N)}{d_N q_T^2}$. In this case, we can conclude that System (1) has no positive equilibrium;

(b) Suppose that $\mu d_N q_T^2 > d_I d_T q_T d_N + \delta_N d_T(\tau q_T + d_T q_N)$. We can easily verify that if $V^* > \hat{V}$, i.e., $F(\hat{V}) > G(\hat{V})$, then $\frac{\mu\beta V^*}{d_S+\beta V^*} > \frac{d_I d_T q_T d_N+\delta_N d_T(\tau q_T+d_T q_N)}{d_N q_T^2}$ and if $V^* < \hat{V}$, i.e., $F(\hat{V}) < G(\hat{V})$, then $\frac{\mu\beta V^*}{d_S+\beta V^*} < \frac{d_I d_T q_T d_N+\delta_N d_T(\tau q_T+d_T q_N)}{d_N q_T^2}$. Thus, System (1) has no positive equilibrium if $F(\hat{V}) < G(\hat{V})$, and System (1) has a unique positive equilibrium if $F(\hat{V}) > G(\hat{V})$, denoted by $E^* = (S^*, I^*, V^*, N^*, T^*, B^*, A^*)$, where

$$S^* = \frac{\mu}{d_S + \beta V^*}, I^* = \frac{d_T}{q_T}, N^* = \frac{\tau q_T + d_T q_N}{q_T d_N},$$
$$B^* = \frac{-(d_B - q_B V^*) + \sqrt{(d_B - q_B V^*)^2 + 4e\eta}}{2e},$$

$$T^* = \frac{q_T}{\delta_T d_T}\Big[\frac{\mu\beta V^*}{d_S+\beta V^*} - \frac{d_I d_T q_T d_N + \delta_N d_T(\tau q_T + d_T q_N)}{d_N q_T^2}\Big], A^* = \frac{f B^*}{\gamma V^* + d_A}.$$

If System (1) has a unique positive equilibrium $E^* = (S^*, I^*, V^*, N^*, T^*, B^*, A^*)$, then the variational matrix of the linearized system around positive equilibrium E^* of System (1) is

$$J(E^*) = \begin{pmatrix} -d_S-\beta V^* & 0 & -\beta S^* & 0 & 0 & 0 & 0 \\ \beta V^* & -\delta_N N^* - \delta_T T^* - d_I & \beta S^* & -\delta_N I^* & -\delta_T I^* & 0 & 0 \\ 0 & p & -c-kA^* & 0 & 0 & 0 & -kV^* \\ 0 & q_N & 0 & -d_N & 0 & 0 & 0 \\ 0 & q_T T^* & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & q_B B^* & 0 & 0 & -\frac{\eta}{B^*} - eB^* & 0 \\ 0 & 0 & -\gamma A^* & 0 & 0 & f & -\gamma V^* - d_A \end{pmatrix}.$$

Denote the eigenvalues of the matrix $J(E^*)$ by $\lambda_i, i = 1, 2, 3, 4, 5, 6, 7$. By extensive calculations yields that

$$|J(E^*)| = -\delta_I d_N q_T I^* T^*(d_S + \beta V^*)\Big[(c\gamma V^* + cd_A + kd_A A^*) \Big(\frac{\eta}{B^*} + eB^*\Big) + kfq_B V^* B^*\Big] < 0.$$

Since the sum of eigenvalues of the variation matrix equals the trace of the matrix, and the product of eigenvalues equals the determinant of the matrix (Ding et al. 2015; Guo et al. 2020), we use the signs of the trace and determinant of the variation matrix to derive some stability information of the positive equilibria. Because

$$\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \lambda_5 + \lambda_6 + \lambda_7 < 0, \text{ and } \lambda_1\lambda_2\lambda_3\lambda_4\lambda_5\lambda_6\lambda_7 < 0,$$

it follows that the variational matrix $J(E^*)$ may have even eigenvalues with non-negative real parts. Thus, the unique positive equilibrium E^* is either locally asymptotically stable, or the dimension of its unstable manifold and center manifold are both even. □

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Data Availability The data used in this study will be made available on GitHub upon acceptance for publication. The data repository will include all datasets, code, and documentation necessary to reproduce the results presented in this paper. A link to the GitHub repository, along with a persistent identifier (DOI), will be provided in the published version of the paper.

Declarations

Conflict of interest The authors declare that they have to known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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